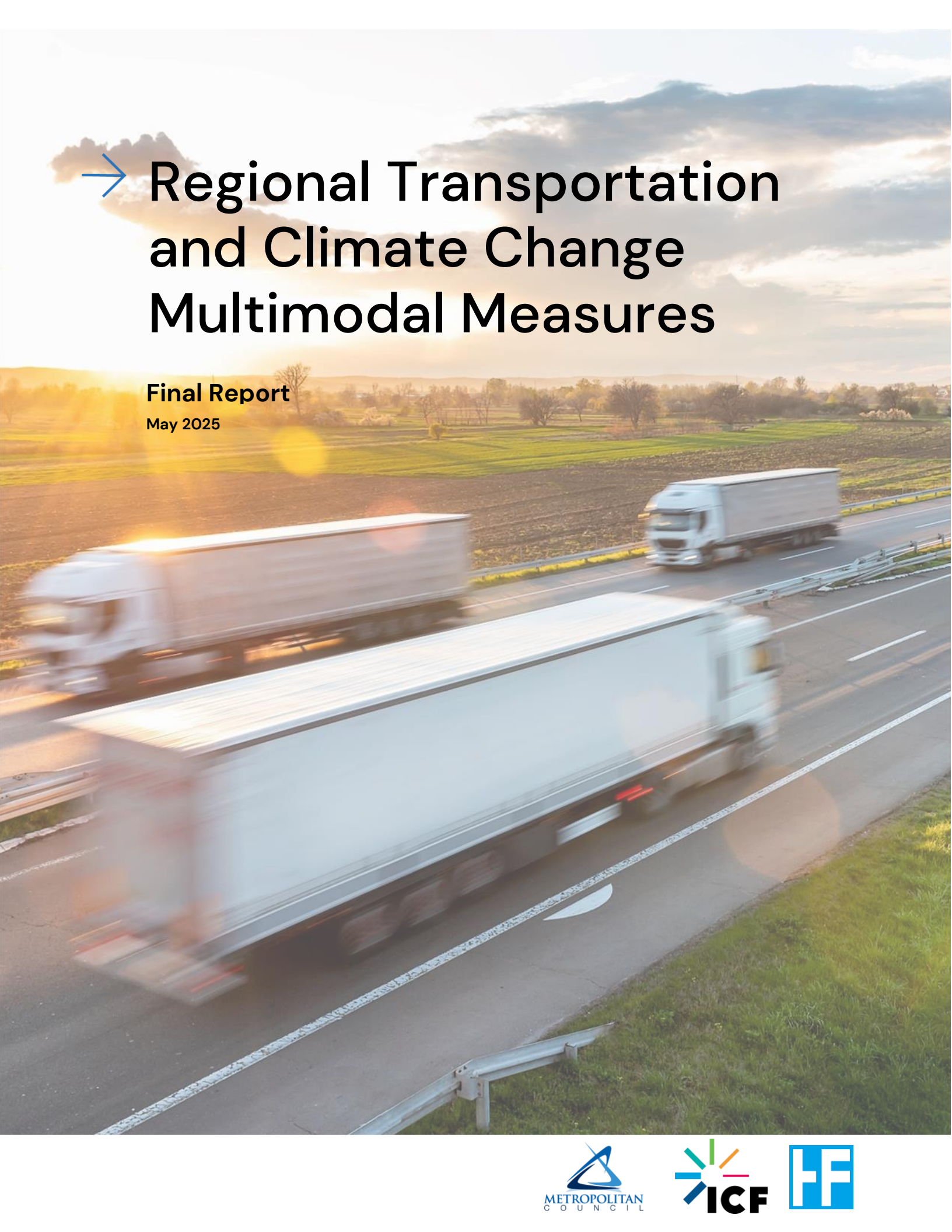




→ Regional Transportation and Climate Change Multimodal Measures

Final Report

May 2025

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Acronym	Definition
AADT	Average Annual Daily Traffic
ABM	Activity-Based Model
AEO	Annual Energy Outlook
BAT	Business Access and Transit
BAU	Business-as-Usual
BEV	Battery Electric Vehicle
Bike/Ped	Bicycle and Pedestrian
BLS	U.S. Bureau of Labor Statistics
BRT	Bus Rapid Transit
CAFE	Corporate Average Fuel Economy
CAPCOA	California Air Pollution Control Officers Association
CARB	California Air Resources Board
CDOT	Colorado Department of Transportation
CMAQ	Congestion Mitigation and Air Quality
CO	Carbon Monoxide
CO ₂ e	Carbon Dioxide Equivalent
Council	Metropolitan Council
CPRG	Climate Pollution Reduction Grants
CTU	Cities, Townships, and Unorganized Territories
DOT	Department of Transportation
DRCOG	Denver Regional Council of Governments
EIA	Energy Information Administration
EPA	Environmental Protection Agency
EV	Electric Vehicle
FHWA	Federal Highway Administration
FTA	Federal Transit Administration
REET	Greenhouse gases, Regulated Emissions, and Energy use in Technologies
HCM	Highway Capacity Manual
HD	Heavy-Duty
HEV	Hybrid Electric Vehicle
ICE	Internal Combustion Engine
kWh	Kilowatt Hour
LD	Light-Duty
MA3T	Market Acceptance of Advanced Automotive Technologies
MassDOT	Massachusetts Department of Transportation
MD	Medium-Duty
MICE	Minnesota Infrastructure Carbon Estimator
MnDOT	Minnesota Department of Transportation
MOVES	Motor Vehicle Emission Simulator
MPCA	Minnesota Pollution Control Agency
MPO	Metropolitan Planning Organization

MT	Metric Ton
MTC	Metropolitan Transportation Commission
MTC	Metropolitan Transportation Commission
NEVI	National Electric Vehicle Infrastructure
NEVI	National Electric Vehicle Infrastructure
NOx	Nitrogen Oxides
NTD	National Transit Database
ODOT	Oregon Department of Transportation
PCAP	Priority Climate Action Plan
PHEV	Plug-in Hybrid Electric Vehicle
PSRC	Puget Sound Regional Council
RMI	Rocky Mountain Institute
RS	Regional Solicitation
SANDAG	San Diego Association of Governments
SCC	Social Cost of Carbon
SHCN	Sustainable Healthy Cities Network
SHIFT	State Highway Induced Frequency of Travel
SMTP	Statewide Multimodal Transportation Plan
SMTP	Statewide Multimodal Transportation Plan
SP	Strategy Planning
STIP	State Transportation Improvement Program
TBI	Travel Behavior Inventory
TDM	Travel Demand Model
TIP	Transportation Improvement Program
TMA	Transit Market Area
TPP	Transportation Policy Plan
USDOT	U.S. Department of Transportation
VMT	Vehicle Miles Traveled
VOC	Volatile Organic Compound
WG	Working Group
WTW	Well-to-Wheels
ZEB	Zero-Emission Bus
ZEV	Zero-Emission Vehicles

Executive Summary

The Twin Cities metropolitan area, centered in the cities of Minneapolis and Saint Paul, is home to 3 million people in 7 counties and 186 cities and townships. The Metropolitan Council (Council) is responsible for making investments in developing bus and rail transitways and provides grants to support multimodal development, which aims to increase transportation choices, expand housing and living options, and encourage economic activity. Given that transportation contributes over 22% of regional GHG emissions, the Council has undertaken this initiative to improve methodologies for evaluating the greenhouse gas (GHG) impacts of transportation projects and investments across all mobility modes, including walking, biking, transit, highways, and electrification. This study also aims to enhance regional planning and investment decisions aligned with climate and sustainability goals.

The key objectives of this initiative include:

- Understand existing procedures to quantify GHG emissions within planning processes and documents, including the Transportation Policy Plan (TPP), Transportation Improvement Program (TIP), and Regional Solicitation (RS).
- Review peer agency practices, academic literature, and available tools for GHG emissions assessment.
- Develop and implement reliable, straightforward, and region-specific methodologies for quantifying GHG emissions from various transportation projects.
- Support the region's commitment to reducing transportation-related emissions and developing a holistic view of the impact of transportation projects.

The study has examined the following planning documents and methodologies:

- 2040 TPP: the Twin Cities metropolitan area's long-term policies and plans to guide the development of the region's transportation system, carrying forward the vision toward economic success and vibrancy in the decades to come.
- 2023–2026 TIP: a four-year funding plan that includes all proposed federally-funded transportation projects within the metropolitan planning area, including highway, transit, bike and pedestrian improvements.
- 2022/2024 RS Applications: a competitive solicitation process to distribute federal funds for roads, bridges, and transit, as well as biking and walking infrastructure.
- Peer agencies' planning documents and project evaluation processes: GHG quantification and project evaluation/prioritization procedures from other state departments of transportation (DOTs) and Metropolitan Planning Organization (MPO).
- Quantification toolkits: publicly available toolkits from the Federal Highway Administration (FHWA), the Environmental Protection Agency (EPA) and traffic simulation software.

The recommendations below are suggested for the Council to consider in upcoming planning processes:

- Standardize the scope of vehicle emissions and leverage tools such as MOtor Vehicle Emission Simulator (MOVES) and Greenhouse gases, Regulated Emissions, and Energy use in Technologies (GREET) to account for vehicle well-to-wheels (WTW) emissions for all fuel types.
- Encourage project sponsors to use the Minnesota Infrastructure Carbon Estimator (MICE) tool to quantify lifecycle GHG emissions relating to infrastructure construction and maintenance.

- Account for the impact of induced vehicle travel (additional vehicle activities resulting from roadway projects) in project evaluation and start systematically monitoring traffic data from funded projects to refine the methodology for calculating induced demand.
- Align electric vehicle (EV) penetration rates with EPA emission standards to accurately forecast future business-as-usual (BAU) forecasts and develop scenarios to evaluate impact of accelerated EV adoptions based on regional specific goals.
- Strike a balance between the efforts necessary to implement GHG quantification methodologies and the accuracy of the results obtained.

The outcome of this study has provided a foundation for the Council to pursue the following directions:

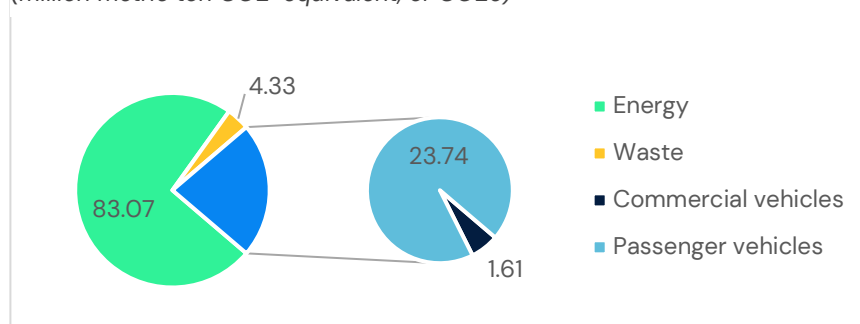
- Integrate the newly developed R-based Transportation Project Emission Reduction Calculator tool into future RS evaluation processes.
- Implement a tiered approach for GHG evaluations, aligning level of effort with project impact.
- Continue collaborating with MnDOT, Minnesota Pollution Control Agency (MPCA), and local partners to gather regional data, improve vehicle technology adoption forecasts, as well as refine and align quantification methodologies for project GHG evaluation and induced demand estimates.

The Council is well-positioned to improve the existing GHG impact assessment and integrate recommended adjustments into their transportation funding and investment prioritization processes. There are significant opportunities for improvement in evaluating GHG impacts, which can subsequently improve the current decision-making process. Continued collaboration, data collection, and refinement of the evaluation processes will be key to achieving a sustainable and resilient transportation future in the Twin Cities area.

Introduction

The Metropolitan Council (Council) is the federally designated Metropolitan Planning Organization (MPO) for the Twin Cities metropolitan area. Centered in the cities of Minneapolis and Saint Paul, the area is made up of the seven counties of Anoka, Carver, Dakota, Hennepin, Ramsey, Scott and Washington. The metro area is home to 3 million people in 7 counties and 186 cities and townships, encompassing nearly 3,000 square miles. The Council makes strategic investments in a growing network of bus and rail transitways, and awards grants to support multimodal development to increase transportation choices, expand housing and living options, and foster economic activity. The Council also partners with local governments to coordinate the orderly development of the metro area, and provides innovative, efficient regional services that benefit from economies of scale.

Figure 1. Greenhouse gas emissions by sectors in the Twin Cities region, 2021 (million metric ton CO₂-equivalent, or CO₂e)



Considering the Council's significant role in prioritizing investments and making informed planning decisions, understanding the comprehensive impact of different transportation projects across all mobility modes is critical. Given the recent increased focus on reducing greenhouse gas (GHG) emissions from transportation sources, which contributes to more than 22% of the region's total GHG

emissions, as shown in **Figure 1** [1], it is important to develop a comprehensive set of methodologies to assess the GHG benefits and costs associated with individual transportation projects and programs.

To facilitate this initiative, the Council has undertaken this work to review, improve and develop methodologies (i.e., data requirements, logic, formulas, off-the-shelf software tools) for evaluating the GHGs impacts of mobility type transportation projects for all modes, including walking, biking, transit, highways transportation demand management, and electrification. As part of the study, the consultant team from ICF and HFTE (hereby referred to as the consultant team) has evaluated peer agencies' GHG evaluation methodologies, reviewed academic and professional literature for best practices and insights for calculating GHGs for different types of projects, developed a categorization scheme for project evaluations, produced a guide for calculating GHGs in the region at project and program scales, and identified the types of projects with the greatest potential for positive and negative impacts on GHGs. This study focused on reviewing transportation projects and refining methodologies for quantifying greenhouse gas emissions in the following documents and processes:

- **2040 Transportation Policy Plan (TPP).** As the region's long-range transportation plan, the TPP identifies the vision, goals, and objectives for the transportation system and the priority transportation investments and programs over a 20-year horizon. The team reviewed the 2040 Transportation Policy Plan (TPP), also referred to as Thrive MSP 2040, as it was the most current TPP available at the time of the study [2]. The TPP reports aggregate 2040 on-road transportation GHG emissions under the three modeled plan scenarios (no build, current revenue, increased revenue).
- **Transportation Improvement Program (TIP).** It is required by federal law for the Council to produce a four-year TIP that outlines funded highway, transit, and bicycle and pedestrian (bike/ped) projects within the metropolitan planning area. In this study, the team focused on the 2023 through 2026 TIP, which

includes projects valued at approximately \$5.6 billion [3]. Currently there is no calculation of GHG emissions impacts of TIP projects.

- **Regional Solicitation (RS).** The RS is a process where the Council prioritizes and invests in roads, bridges, transit, biking, and walking needs every two years, awarding about \$250 million in federal funds. The team has reviewed both 2022 and 2024 RS applications for this study. RS currently provides detailed guidance for scoring projects across six to nine criteria, depending on the project category [4]. For projects that expand roadway capacity or improve traffic flow, the calculation of emissions impacts depends on the change in vehicle speed, as predicted using a Synchro, or Highway Capacity Manual/Highway Capacity Software (HCM/HCS) analysis. The scoring guidance calls for calculating the sum of criteria pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x), and volatile organic compound (VOC) emissions—GHG emissions are not currently considered. For transit and transportation demand management projects, the emissions impacts are determined based on the calculated reduction in vehicle miles travelled (VMT).

The evaluation of current methods and recommendations also considered well each method addresses elements and issues of interest to the Council, such as direct vs. indirect impacts, construction impacts, and lifecycle analysis. The scope and boundaries of GHG quantification evaluated in this study include:

- **Vehicle tailpipe vs. well-to-wheels (WTW) emissions.** A comprehensive GHG analysis will consider both emissions from the vehicle tailpipe as well as the emissions associated with production of the fuel (often referred to as “upstream” emissions), including electricity generation. Some methods developed for traditional air quality analyses consider tailpipe emission only.
- **Project construction vs. system user emissions.** Quantifying the emissions associated with project construction and maintenance requires entirely different methods than system user impact emissions. In some cases, construction and maintenance will be negligible and can be ignored.
- **Embodied materials emissions.** Analogous to the upstream emissions associated with vehicle fuels, construction project analysis can consider the upstream emissions associated with project materials such as cement, asphalt, steel, etc.
- **Vehicle manufacturing and disposal.** A comprehensive lifecycle GHG analysis of transportation investments may also include emissions associated with vehicle manufacturing and disposal. However, because these elements are outside the control of transportation agencies and because they introduce significantly more complexity and uncertainty into an emissions analysis, it is often recommended that they be excluded.
- **Induced vehicle travel.** It is well documented that roadway capacity expansions will induce new vehicle travel. These effects include short-term elements (e.g., mode shift to auto, longer auto trips) and longer-term elements (e.g., more dispersed land development), and should be examined carefully in transportation planning projects.

The significance and complexity of each component was analyzed for various project categories and based on the review of best practices from other agencies nationwide, the consultant team has provided recommendations for TPP, TIP, and RS GHG quantification respectively. The primary objective of these recommendations is to equip the Council with reliable methodologies to assess the GHG impact of projects, while minimizing the efforts and staff resources associated with implementing these calculations. This work has also been done in consultation with the Minnesota Department of Transportation (MnDOT), the Minnesota Pollution Control Agency (MPCA), Council technical committees, and other stakeholders.

Inventory and Evaluate Current Methods

The consultant team reviewed the methodologies for calculating GHG emissions in the 2040 TPP, 2023–2026 TIP, and 2022/2024 RS projects. This summary outlines the current capabilities and gaps observed in each planning process.

2040 Transportation Policy Plan

The TPP GHG emissions projections make use of analyses conducted with the MOtor Vehicle Emission Simulator (MOVES) and the regional travel demand model (TDM). MOVES is a state-of-the-science emission modeling system that estimates emissions for mobile sources at the national, county, and project level for criteria air pollutants, greenhouse gases, and air toxics, managed and maintained by the Environmental Protection Agency (EPA) [5].

The regional TDM is used to forecast future travel demand using projected changes to population, jobs, the transportation network, and land use. It considers the trips of users across the entire system and contains a transit network and also a mode choice sub-model for non-motorized trips including bicycle and pedestrian trips. The model is effective at modeling different geographies due to its use of a model of the actual transportation network as well as different land use types and densities. Due to the activity-based nature of the TDM, changes to level of service of one mode can lead to increases or reductions of usage in another. For example, increased transit service could lead to less congestion on a roadway, leading to an increase in vehicles using that roadway for their trajectory. The iterative approach to these models enables an equilibrium to be reached in mode choice and congestion that does reflect the overlapping nature of the various modes of transportation. The methodologies used to inform the TPP cannot produce estimates of the component parts of GHGs, especially as it pertains to construction, manufacturing, and disposal. Emissions are limited to those due to system user impacts.

Assuming input data is already gathered and specified for the models, these tools can take several hours to several days per scenario to run and generate raw outputs. Additional data aggregation is required following model completion to translate VMT and emissions factors into reportable figures. These tools could be classified as a high level of effort.

2023–26 Transportation Improvement Program

The inventory of TIP (Table 1) selected projects is limited to project descriptions and awarded dollars. As anticipated, there is limited information to support the calculation of changes in GHG emissions, so a detailed evaluation was not performed.

Table 1. 2023–26 TIP Project Types

Project Type		Description of Project Type	Grand Total
Appurtenance	Education and Safety	Travel behavior inventory and regional model development	\$3,166,000
	Historic Preservation	Study for Rondo Area Street Improvements in St Paul	\$1,400,000
	Median Barrier	Cable median barrier installation and resurfacing/repair projects	\$12,342,000
Bike/Ped	Bridge Pedestrian	Construction and replacement of pedestrian bridges	\$9,746,640
	Coop Const Agreement	Construction of bike facility	\$2,670,000
	New Trail	Construct bike/ped trail or bridge	\$37,147,057

Project Type		Description of Project Type	Grand Total
	Pedestrian Ramps (ADA Improvements)	Construct ped safety improvements, protected bikeways and multiuse trails	\$52,945,442
Bridge Replacement or Construction	Bridge New	Construction of bridge	\$6,050,000
	Bridge Removal	Removal of bridge	\$2,574,064
	Bridge Replacement	Rehabilitation or replacement of bridges, interchanges	\$53,755,895
Bridge Rehabilitation	Bridge Deck Overlay	Rehabilitation of bridges	\$109,421,000
	Bridge Repair	Repair bridges	\$48,609,520
Grade and Resurface	Auxiliary Lane	Involves construction of auxiliary lanes	\$11,307,000
	Coop Const Agreement	New frontage roads or connections	\$2,912,000
	Major Construction	Interchange construction	\$879,003,154
Pavement Resurface and Rehabilitation	Bituminous Overlay	Resurfacing projects	\$6,998,734
	Concrete Overlay	Highway resurfacing projects	\$169,021,000
	Concrete Pavement Rehab	Highway resurfacing and repair projects	\$85,574,000
	Mill and Overlay	Resurfacing and reconstruction projects	\$60,194,000
Traffic Control Devices/Safety	Channelization	(Re)construction of reduced conflict intersections and lane expansion projects	\$43,464,300
	Other	Reconstruct sidewalks, improve crossing, reconfigure lanes	\$8,592,500
	Pavement Markings	Convert 4-lane section to 3-lane section, mill and overlay	\$2,168,226
	Roundabout	Construct roundabout projects	\$17,414,223
	Turn Lanes	Reconstruct roads and turn lanes	\$5,877,667
Transit	Transit Grant Capital Improvement (Non-vehicle)	(Implied)	\$2,034,300,769
	Transit Operations	(Implied)	\$145,231,823
	Transit Vehicle Purchase	(Implied)	\$122,561,895

2022/24 Regional Solicitation

Presently, many RS projects rely on near-term vehicle volumes or anticipated near-term new usage (in the case of transit or bike/pedestrian infrastructure projects). The RS process does not consider shifting trends in use over the long term and corresponding cumulative emissions over these years. In order to implement a longer-term estimate of emissions, updated methodologies would require information regarding land use, population, and demographic projections which could add considerable complexity.

The recommended methodologies in the 2022/24 RS sample applications range from single calculations and spreadsheet analyses to the extraction of projected VMT from existing completed regional model runs and new analyses results from Synchro or an HCM software suite. These methodologies require between a low to medium level of effort to perform.

One thing to note is there are no project types that consider electric vehicles (EVs) except for Unique Projects, under which EV charging stations can be considered. The method for determining the mitigation

impact of these stations is left up to the applicant and does not specify to consider a changing electricity generation mix. **Due to the increasing applications related to EVs and the installation of charging infrastructure in the Twin Cities region, it may be necessary to establish a new project category with a well-defined methodology focused on EVs.**

Summary of Current Methods

Table 2 outlines the methods for estimating GHG emissions in the current TPP, TIP, and RS processes, differentiating between direct and indirect GHG impacts. Direct GHG impacts include construction, induced vehicle purchases, fuel consumption, traffic speed/congestion, changes in stop/start frequency, etc. Indirect GHG impacts are related to induced demand, such as the use of targeted and other travel modes, and land use changes that subsequently affect transportation-related GHG emissions.

Table 2. Summary of how different GHG impacts are being addressed in the current TPP, TIP, and RS planning processes

GHG Impact Scope		TPP	TIP	RS
Direct GHG Impacts	Construction	N/A	N/A	N/A
	Induced Vehicle Purchase	Vehicle type distributions are held constant throughout future years according to the percentage of vehicle types from 2015 vehicle registration data	N/A	N/A
	Fuel Used	MOVES county-level generic inputs are used. GHGs from fuel use is estimated using estimated fleet mix, modeled VMT and MOVES emissions outputs	N/A	- Several non-highway projects, such as Transit Expansion, Arterial Bus Rapid Transit (BRT), Transit Modernization, and transportation demand management specify an emissions factor of 366.60g CO₂e / mi for avoided VMT from personal automobiles. - Roadway projects, including traffic management, spot mobility and safety, and strategic capacity use a more nuanced approach involving total emissions reported from Synchro or HCM software studies.
	Impacts on Speed of Traffic/ Congestion	The TDM uses an iterative approach to balance volume and delay. Each road has a volume-delay function. Links that are congested affect trip behavior, including forecasted daily activity patterns, destination choices, and mode choice.	N/A	Roadway projects that use Synchro or HCM consider congestion and vehicle speeds when calculating emissions outputs.

	Change in Frequency of Stops/ Starts	MOVES models stops and starts. The default behavior is used.	N/A	Roadway projects that use Synchro or HCM consider stops and starts when calculating emissions outputs.
Indirect GHG impacts	Induced Demand	The TDM uses an iterative approach to balance volume and delay accounts for induced demand.	N/A	N/A
	Land Use	The TDM does not use the transportation system to inform future changes to land use. Land use projections are treated as a snapshot and are used to inform travel demand.	N/A	N/A
	Mode Shift	The TDM uses a mode choice model to determine the optimal mode to be used for each simulated individual's day of trips	N/A	• Traffic Management Technologies and Strategic Capacity project types offer methods to quantify the present and forecasted traffic volumes from the regional model. • Some Transit projects use mode shift and assume that every rider is being shifted from personal vehicle. • Transportation demand management projects assume that VMT avoided is the product of total number of one-way commute trips avoided times a flat one-way commute trip length of 12.1 miles.

Review and Summarize Best Practices

After examining the existing methodologies and scope, the consultant team then conducted a thorough review of current best practices among other agencies for calculating the GHG impacts of transportation projects, investments, and other strategies. The review focuses on the types of projects and strategies of interest to the Council – those that could be included in the TPP, TIP, and RS. The best practices described in this section were also used to develop recommendations for improvements to the Council’s analysis methods.

Project-Level Best Practices

The best practices to quantify GHG emissions at project-level were identified by reviewing documents and analysis tools developed by other public agencies that quantify the GHG impacts relevant transportation projects, programs, and strategies. The results of this best practice review are organized by category and type of project, as shown below.

Table 3. Project-level GHG methodologies reviewed in this study.

Modal Category	Project Category	Project Type
Roadway	Traffic Flow Improvements	Intersection Improvements
		Traffic Signal Synchronization
		Roundabouts
		Mid-Block Two-Way Left Turn Lanes
	Managed Lanes	New Managed Lane Facilities
		Managed Lane Conversions
Transit	Transit Improvements	New or Expanded Transit Service/Facility
		Transit Frequency Improvement
		Transit-Supportive Roadway Treatment
		Bus Rapid Transit Project
		Transit Fare Reduction
Travel Demand Management	Travel Demand Management	Bikeshare
		Carshare
		Rideshare
		Parking Management
		Telework Strategies
		Employer Commute Trip Reduction Programs
		Community-Based Travel Planning/ Marketing
Bicycle and Pedestrian	Multiuse Trails and Bicycle Facilities	Bicycle Path or Lane
		Bicycle Boulevard
	Pedestrian Facilities	Pedestrian Facility Improvement
Electrification	Government Fleet EV Purchasing	
	EV Carsharing	

	Public EV Charging Infrastructure
	EV Promotion and Marketing Campaigns

California Climate Investments – Calculator Tools

The California Air Resources Board (CARB) implements a number of GHG reduction funding programs under the umbrella of “California Climate Investments” [6]. Most of these programs require that applicants estimate emission reductions as part of the application process. CARB has created calculator tools and accompanying quantification methodology reports to assist applicants with estimating emission reductions.

One program, the Sustainable Transportation Equity Project, includes three types of strategies that are eligible for funding:

- **Active Transportation.** Includes projects such as bicycle, pedestrian, and complete streets infrastructure (e.g., crosswalks, sidewalks, bikeways).
- **Fixed-route Transit.** Includes projects such as transit station improvements, transit right-of-way improvements (e.g., bus rapid transit lanes), transit operations improvements, transit passes, and improvements to increase accessibility of transit (e.g., network/fare integration, wayfinding, and signage).
- **Shared Mobility.** Includes projects such as zero-emission carshare, rideshare, vanpooling, carpooling, bikeshare, scooter share, ride-hailing, shuttles, and microtransit (on-demand transit) services.

Another program, Clean Mobility Options, can provide funding for the following types of strategies:

- **Shared Use Mobility.** Includes carshare, bikeshare, and scooter-share.
- **Ridesharing.** Includes carpool and vanpool.
- **Public Transit.** Includes “innovative transit” (e.g., on-demand shuttles and circulators, paratransit services, microtransit) ride on-demand (taxi service and transportation network companies), and traditional public transit (e.g., fixed-route bus service).

Some of the CARB formulas are derived from methods developed previously and used to estimate emissions benefits of air quality control measures, as documented in CARB’s “Methods to Find the Cost-Effectiveness of Funding Air Quality Projects” dated 2005.

Handbook for Analyzing Greenhouse Gas Emission Reductions, Assessing Climate Vulnerabilities, and Advancing Health and Equity

The California Air Pollution Control Officers Association (CAPCOA) is made up of representatives from the state’s local air pollution control districts. CAPCOA initially sponsored the development of a GHG analysis handbook in 2010, and the handbook was significantly expanded and updated in 2022 [7]. The handbook is used by local governments and others across California to estimate reductions in GHG emissions from new development projects and to create climate action plans, master plans, and general plans. The handbook contains methods for 30 transportation measures, grouped in the following six categories:

- Land Use
- Trip Reduction Programs
- Parking or Road Pricing / Management
- Neighborhood Design
- Transit
- Clean Vehicles and Fuels

Note that the methodologies and formulas in the handbook calculate a percent reduction in GHG emissions relative to a baseline project without the measure, not an absolute magnitude of GHG emissions.

CMAQ Emissions Calculator Toolkit

The Congestion Mitigation and Air Quality (CMAQ) program provides funding to state and local governments for transportation projects and programs that can reduce emissions. The Federal Highway Administration (FHWA) has created the CMAQ emissions calculator toolkit as a resource for agencies to estimate emissions reductions for CMAQ projects and help meet annual reporting requirements [8]. The toolkit includes 13 tools, covering more than 20 project types, including transit service expansion, alternative fuels, and bicycle and pedestrian improvements.

Mobility Management Toolbox – VMT Reduction Calculator Tool

The San Diego Association of Governments (SANDAG) sponsored the development of the Mobility Management Toolbox to assist local governments with identification and analysis of transportation demand strategies [9]. The Toolbox includes an Excel-based calculator tool for estimating strategy VMT impacts. The VMT tool includes 22 strategies grouped in the following five categories:

- Employer Commute Programs (primarily employer-based programs)
- Land Use Strategies (transit-oriented development, mixed-use development)
- Parking Management (parking pricing and parking cash-out)
- Neighborhood Enhancements (e.g., pedestrian facilities, bicycle facilities, carshare, bikeshare)
- Transit Strategies

Project Level Emissions Estimation Tool

The Puget Sound Regional Council (PSRC) sponsored the development of this tool in 2015 for the primary purpose of calculating the emissions impacts of CMAQ projects. The Excel-based tool facilitates analysis of 13 strategy types, including the following:

- New or Expanded Transit Service
- Transit Station Amenities
- Business Access and Transit (BAT) Lanes
- New or Upgraded Bicycle/pedestrian Facility
- Reduce Intersection Delay
- Increase Corridor Speed
- Outreach Programs & Subsidized Transit
- Vanpools

Minnesota Infrastructure Carbon Estimator (MICE)

MICE is a tool that calculates the lifecycle energy and GHG emissions related to the construction and maintenance of transportation infrastructure [10]. MICE is based on the Infrastructure Carbon Estimator (ICE) tool, which was originally developed by FHWA; MICE has been customized for use in Minnesota by MnDOT. The tool facilitates a planning-level estimation of embodied carbon emissions in transportation facilities without requiring engineering studies. The tool is equipped to evaluate emissions from a range of infrastructure, including bridges/overpasses, roadways, bus rapid transit, parking lots, bike and pedestrian facilities, roadway lighting and signage, light and heavy rail, vehicle operations, and custom pavement. Unlike other sources reviewed for this document, the emissions estimates from MICE account for the “upstream” activities associated with material extraction, production, and transport – not just the tailpipe emissions from motor vehicles.

The project-level GHG quantification methodologies were captured and recorded, with specific variables, value ranges, units, and data sources identified where feasible. For each project category and subcategory, this report summarizes the best practices for emissions impact analyses. The methods are organized by category, project type, and source, so that project types with more than one reviewed methodology can be reported adjacent to each other for future selection and recommendation. The matrix below in Table 4 illustrates which project types are included in each of the key information sources.

Table 4. Project types included in the key information sources.

Project Type	CARB	CAPCOA	FHWA	SANDAG	PSRC	MnDOT	Other
Intersection Improvements			●		●		
Traffic Signal Synchronization			●				
Roundabouts			●				
Mid-Block Two-Way Left Turn Lanes			●				
New Managed Lane Facilities			●				
Managed Lane Conversions			●				
New or Expanded Transit Service/Facility	●	●	●	●	●		
Transit Frequency Improvement	●	●	●	●			
Transit-Supportive Roadway Treatment		●		●			
Bus Rapid Transit Project	●	●					
Transit Fare Reduction		●		●			
Bikeshare	●	●		●			
Carshare	●	●		●			●
Rideshare		●	●	●			
Parking Management		●		●			
Telework Strategies				●			
Employer Commute Trip Reduction Programs		●		●			
Community-Based Travel Planning/ Marketing		●		●			
Bicycle Path or Lane	●	●	●	●	●		
Bicycle Boulevard		●					
Pedestrian Facility Improvement	●		●		●		
Electrification Projects		●					●
Project Construction						●	

Programmatic and Plan-Level Analysis

This study primarily addresses methods to evaluate emissions impacts of individual strategies or investments, often done at the project-level. However, many agencies including the Council have developed tools and methods to estimate the impacts of bundles of investments or strategies at a programmatic or plan scale and at higher-level geographies including states, regions, municipalities, and corridors. These methods are often applied as part of a regional planning process. A handful of transportation agencies have also estimated GHG emissions impacts from TIP. Traffic modeling software is used for estimating travel and emissions impacts of corridor-scale improvements. This section reviews these initiatives and similar ones by peer agencies such as the state DOTs, SANDAG, and the Metropolitan Transportation Commission (MTC), the MPO for the San Francisco Bay Area.

Transportation Improvement Programs

SANDAG Regional Transportation Improvement Program

The 2022 Regional Transportation Improvement Program (RTIP) for the San Diego region prioritizes the continuation of projects from previous State Transportation Improvement Program (STIP) periods, including significant initiatives like the SR 94/ SR 125 Interchange, Arterial Operational Improvements Design, and the I-5 North Coast Corridor Project (I-5 NCC) [11]. The I-5 NCC project alone encompasses a wide array of components, including HOV lanes, bridge widening, soundwalls, bike trails, and more.

The proposed 2022 STIP incorporates the following aspects related to greenhouse GHG emissions:

- **Project-Specific GHG Evaluation:** Individual projects within the STIP include a specific indication of whether they will "Reduce Greenhouse Gas Emissions," with a "yes" or "no" checkbox. This simplistic approach to identifying projects' potential to reduce emissions provides a high-level understanding of how each project aligns with GHG reduction goals.
- **Lack of Detailed Methodology:** While the STIP acknowledges the relevance of GHG emissions, it does not provide clear information about the methodology used for these assessments. It remains unclear whether these are broad estimations or detailed analyses comparing the build and no-build alternatives for each project.

In conclusion, SANDAG's 2022 STIP demonstrates an awareness of the importance of GHG emissions in transportation projects, reflecting this through a straightforward indication for each project. However, the lack of transparency regarding the methodology or the detailed analysis of GHG impacts suggests that a more comprehensive and transparent approach may be needed to thoroughly assess and report on the greenhouse gas implications of the projects within SANDAG's 2022 STIP.

Oregon DOT STIP

The STIP is Oregon DOT's (ODOT) capital improvement plan for state and federally-funded projects. Transportation projects large and small can impact GHG emissions, by creating more emissions from vehicles traveling on the system, by reducing emissions from vehicles, or having little effect either way. ODOT looked at ODOT-led and -administered projects for:

- **User emissions**, which are generated through drivers using the transportation system through 2050. This includes vehicle tailpipe emissions and fuel production emissions (such as electricity produced elsewhere). These represent the majority of transportation sector emissions.
- **Embodied emissions**, which are generated during the construction of transportation projects. These include emissions from materials production, delivery, and construction.

- **Funding impacts**, which shows dollars spent on STIP projects (independent of the magnitude of emissions impacts) that advance GHG emission reduction goals, challenge those goals, or maintain the status quo by having little effect either way. This simple accounting method allows ODOT to consider funding from the full STIP and easily compare to the prior 2021–24 STIP.

Review of the ODOT STIP revealed that the analysis was conducted using [VisionEval](#), a stand-alone R-based strategic modeling tool [12].

Massachusetts DOT TIP/STIP

Massachusetts DOT (MassDOT) Office of Transportation Planning provides a guidance document to ensure MassDOT and MPOs assess and report project GHG emission impacts in a consistent manner.

The 2020 TIP GHG assessment and reporting guidance provided by MassDOT recommended the use of CMAQ spreadsheet calculators and the use of emissions factors for the following project types:

- Bicycle and pedestrian infrastructure
- Bus replacement
- New/additional transit service
- Park and Ride lot
- Traffic operational improvement
- Complete Streets projects
- Alternative fuel vehicles
- Anti-idling strategies
- Bike sharing projects
- Induced travel
- Speed reduction programs
- Transit signal priority
- Truck stop electrification

All TIP projects are subject to a GHG assessment and MassDOT will provide the appropriate CMAQ spreadsheet to estimate the CO₂ impact number, at a minimum.

Colorado DOT STIP

The Colorado Department of Transportation's (CDOT) Regulation Governing Statewide Transportation Planning Process and Transportation Planning Regions (2 CCR 601–22, known as the Greenhouse Gas Transportation Planning Standard), adopted in December 2021, requires the Denver region to reduce surface transportation GHG emissions through the transportation planning process. The regulation (2 CCR 601–22, Section 8.02.6) establishes GHG emission reduction levels from an established baseline for four analysis years: 2025, 2030, 2040 and 2050. For the Denver region, baseline GHG values are established based on the defined transportation investments and planning assumptions identified in the 2050 RTP adopted in April 2021. The target GHG emissions are determined by subtracting the rule's GHG emission reduction levels from the total baseline emissions for each analysis year.

According to the Denver Regional Council of Governments (DRCOG) Transportation Greenhouse Gas Report, CDOT models GHG emissions using a consolidation of transportation demand, travel demand, and emissions modeling [13]. For estimating demands of the transportation system over time, CDOT leverages the [UrbanSim](#) model, which simulates household and other sociodemographic qualities to anticipate the effects on the transportation network; for simulating the planned transportation system and land use attributes that affect trips to the roadway and transit network, CDOT uses an activity-based TDM from the DRCOG; for estimating transportation emissions, CDOT leverages MOVES model, adjusted for the state of Colorado based on MOVES region-inputs as well as the regional TDM output.

Long-Range Regional Plan Analysis

The MTC is the MPO for California's San Francisco Bay Area. Under California law, all California MPOs are required to demonstrate that their long-range transportation plans will achieve per-capita light-duty vehicle GHG emission reduction targets set by CARB. For example, the CARB target that was in place for MTC for the

2017 regional transportation plan was a 7% reduction for 2020 and a 15% reduction for 2035, as compared to 2005 GHG levels. Importantly, MTC and the other California MPOs cannot “take credit” for emission reductions that result from state or federal actions, such as the state’s zero-emission vehicle (ZEV) mandates or federal Corporate Average Fuel Economy (CAFE) standards; the emission reductions must result from investments, programs, and policies implemented by MTC or other regional partners.

Like most large MPOs in California, MTC has demonstrated compliance with the CARB GHG reduction targets through two main approaches:

- Running the regional TDM and applying emission factors to the vehicle travel outputs;
- “Off-model” calculations, typically using Excel spreadsheet analyses.

MTC’s Travel Model Analysis

MTC’s TDM is used to capture the impacts of major changes to the regional transportation system, forecasted changes to land use (population and employment growth), and major policy changes (e.g., region-wide road pricing). Like many MPOs, MTC analyzes several scenarios as part of its long-range planning process. For example, the table below shows that, for the *Plan Bay Area 2040* adopted in 2017, in addition to the Proposed Plan scenario, MTC also analyzed “Main Streets”, “Big Cities”, and “Environment, Equity, and Jobs” scenarios [14].

Table 5. MTC modeled scenarios for Plan Bay Area 2040.

Alternative	2005	2015	2020	2030	2035	2040
Historical	●	●				
No Project			●		●	●
Proposed Plan			●	●	●	●
Main Streets			●		●	●
Big Cities			●		●	●
Environment, Equity, and Jobs			●		●	●

Examples of investments in the Proposed Plan scenario that are assumed to be captured by the travel model analysis include:

- Extension of the BART heavy rail system to Silicon Valley
- Extension of the Caltrain commuter rail to downtown San Francisco
- New ferry service
- New bus rapid transit (BRT) corridors
- Light rail transit extensions

The Proposed Plan also included a variety of highway projects, including capacity expansion projects and operational improvements; the travel model is the basis for determining the emissions impacts of these projects.

MTC’s Off-Model Analysis

For the *Plan Bay Area 2040* adopted in 2017, MTC’s travel model analysis did not result in sufficient GHG emission reduction necessary to achieve the targets set by CARB. Therefore, MTC developed additional investments, policies, and programs to reduce GHG emissions and analyzed these strategies outside the travel model. MTC modeling staff determined that the travel model would not be able to capture the impacts of these strategies.

The following strategies were analyzed for their GHG impact using off-model approaches, typically involving spreadsheet analysis.

- **Commuter benefits ordinance** – A regional policy that requires employers with 50 or more full-time employees in the Bay Area to offer their employees incentives to commute to work by modes other than driving alone.
- **Car sharing expansion** – Expansion of car sharing services in selected urbanized areas.
- **Bike share expansion** – Expansion of the region's existing bike share service.
- **Bicycle infrastructure** – Expansion of the region's network of bikeways.
- **Vanpools and employer shuttles** – Implementation of regional vanpool services and employer-operated shuttles.
- **Targeted transportation alternatives** – Implementation of outreach programs that employ a variety of strategies, including individual travel consultation, organized events, and distribution of outreach and informational materials to encourage people to shift from driving alone to carpooling, transit, biking, or walking for any of their trips.
- **Trip caps** – Requirements on new commercial development that sets limits on the number of vehicle trips to and from workplaces, and enforces these limits via regular traffic counts and penalties for non-complying workplaces.
- **Regional electric vehicle charger network** – Expansion of the region's network of electric vehicle charging infrastructure.
- **Vehicle buyback and PEV incentive** – Implementation of a new program that provides a combination of an incentive of up to \$2,500 to purchase a PEV along with the buyback of older, less efficient vehicles.
- **Clean vehicles feebate program** – Implementation of a new feebate program that uses a combination of fees and rebates to change consumer behavior. Consumers purchasing a vehicle that emits more GHG emissions on a gram per mile basis than a defined standard are assessed a fee at the point of purchase. These fees are used to provide rebates to consumers that purchase vehicles that emit less GHG emissions on a gram per mile basis than the defined standard.)

GHG Strategy Planning Tool

The Council is developing a web-based GHG Strategy Planning (SP) Tool to enable municipalities within its jurisdiction to produce GHG emissions forecasts based on baseline data and scenario input assumptions [15]. The Tool relies on data from a variety of sources, including local data sources. The tool also incorporates data and basic assumptions related to residential energy use, the Council's TDM and Travel Behavior Inventory (TBI), the National Transit Database (NTD), and the Energy Information Administration (EIA)'s Annual Energy Outlook (AEO). While the tool integrates many local data sources specific to the metropolitan area, its data and methodology can also be modified within the tool's open-source R code. Importantly, the tool is not a model simulating future scenarios; rather, it produces forecasts based on baseline data and scenario input assumptions.

The Tool considers GHG mitigation strategies in modules related to three sectors: building energy, transportation, and land use/green infrastructure. Synergies between strategies within and across sectors are also considered. Each sector module includes pre-loaded GHG mitigation strategies. For example, mitigation strategies associated with the building energy sector include the percentage of homes that are retrofitted for energy efficiency, the percentage of the electric grid that is decarbonized, the percentage of homes using smart technology, and the percentage of single-family construction replaced with multifamily construction.

Users can create custom scenarios that combine parameters associated with each mitigation strategy, or else use preset scenarios embedded into the tool.

Once a user makes scenario selections, the tool outputs the change in total emissions (in metric tons or MT of CO₂e) between 2018 and 2040, comparing a business-as-usual case to the results from the tested scenario. The tool also provides the reduction impact associated with each sector's strategies such that users can see their relative impact.

Transportation Sector Strategies in the GHG Strategy Planning Tool

Mitigation strategies featured in the transportation module of the Tool focus on reducing transportation sector GHG emissions from two lenses: (1) VMT reduction, and (2) more efficient driving. The module produces 2040 GHG reduction forecasts based on the Tool's baseline data on mode shares, vehicle technologies (e.g., gasoline, diesel, hybrid EVs, plug-in hybrid EVs, etc.), fuel efficiency, and well-to-wheel global warming potential factors per fuel quantity value.

GHG reduction strategies associated with reducing VMT are related to changes in compact development, telework, and road pricing. GHG reduction strategies associated with driving efficiency include those related to vehicle electrification and transit travel. The table below summarizes these strategy categories further.

Table 6. Transportation sector strategy categories – GHG SP Tool¹

Transportation GHG Reduction Lens	Mitigation Strategy Category	Description
VMT Reduction	Compact Development	More compact development can shorten travel distances and make non-personal vehicle travel (e.g., walking, transit) more viable, reducing reliance on personal vehicles and their associated VMT. Examples of actions in this category include increased multifamily zoning, mixed-used development, or transit-oriented development.
	Telework	Telework policies can help reduce commute travel, decreasing VMT due to a reduction of drivers commuting to their workplace.
	Road Pricing	Road pricing actions, such as pay-per-mile car insurance, VMT fees, or gas taxes, can be used to disincentivize driving, subsequently reducing VMT and associated GHGs.
Efficient Driving	Electric Vehicles (EVs)	EVs produce no tailpipe emissions, reducing the GHGs associated with their use. Their overall emissions may be substantially lower than conventional internal combustion engine vehicles depending on the source of electricity used for charging.
	Transit	Transit actions such as increased frequency or coverage, along with increased transit ridership, can help move travelers more efficiently than private vehicle trips. Because of this, heightened transit service or ridership can act to reduce the GHG emissions from private vehicle automobile trips by shifting users to a more efficient mode of travel.

City of Minneapolis' Application of the GHG Strategy Planning Tool

The City of Minneapolis applied the GHG SP Tool to understand the GHG reduction impacts of various scenarios. In the transportation sector, the City examined reductions associated with changes in density,

¹ Adapted from "Climate Planning and Transportation – Empowering Local Governments: Unveiling the Greenhouse Gas Strategy Planning for Sustainable Transportation Solution"

vehicle electrification, transit, telework, and road pricing. A selection of individual strategy results is summarized in the table below.

Table 7. Summary of the City of Minneapolis' transportation sector results

Scenario Strategy	Emissions (thousands of metric tons of CO ₂ e) for Passenger Light-Duty Vehicles			% Change from 2018 Baseline – Business-as- Usual	% Change from 2018 Baseline – Scenario
	2018 Baseline	2040 Business- as-Usual Forecast	2040 Scenario Strategy Forecast		
50% Increase in Population Density	1,070	884	792	17%	-26%
50% Increase in Employment Density			867		-19%
60% vehicle sales battery electric			874		-18%
60% increase in transit service			850		-21%
60% increase in people teleworking			857		-20%
\$1 per mile congestion price			821		-23%
\$50 parking per hour			833		-22%

Traffic Modeling and Simulation Software

Traffic modeling and simulation software is widely utilized by various agencies to assess the emissions impacts of transportation investments and strategies at project and program levels. These software tools play a crucial role in enabling transportation planners and analysts to evaluate the potential environmental effects of different transportation projects and policies. By utilizing inputs such as traffic data, road network characteristics, and vehicle emission factors, these software tools simulate traffic patterns and predict emissions.

Microsimulation and Highway Capacity Manual (HCM) Tools

Many agencies use traffic modeling and simulation software as a standalone tool to estimate emissions impacts alongside traffic impacts. Some commonly utilized modeling tools for this purpose include CORSIM, VISSIM, Synchro/SimTraffic, HCM, and others. Microsimulation tools such as CORSIM, VISSIM, and Synchro/SimTraffic offer direct emission outputs for common pollutants such as CO, NO_x, and VOCs during simulation durations. However, particulate matter or lead outputs are not available in these tools. These outputs can be further used to estimate emissions for specific time frames, such as daily, yearly or project evaluation time periods. It is important to note that while CO, NO_x, and VOCs are air pollutants with various environmental and health impacts, they are not considered GHGs. Users may rely on formulas based on their outputs (VMTs, speeds & stops) from the HCM method to estimate GHG emissions in situations where direct outputs are not available or for validation purposes. By analyzing simulated traffic scenarios, agencies can effectively evaluate the environmental consequences of proposed transportation projects or strategies, enabling them to make informed decisions.

Off-Model Analysis

Moreover, traffic modeling and simulation software is often utilized in conjunction with other tools or formulas, incorporating additional inputs to perform off-model estimates of emissions impacts. These additional inputs may include data on fuel consumption, vehicle types, and travel demand. By integrating these inputs into the modeling process, agencies can enhance the accuracy of emissions estimates and evaluate a broader range of strategies and investments. This approach commonly involves utilizing TDM and the MOVES model for emissions analysis.

Furthermore, some agencies have developed their own tools specifically for estimating emissions in projects requiring GHG analysis. These tools are typically developed in Excel and require general inputs such as traffic volumes, signal timings, and lane miles for different evaluation scenarios. The emission outputs can be at various levels, including project, program planning, on-road, construction, maintenance stages, or even life-cycle assessments. Examples of such tools include the FHWA CMAQ toolkits as described earlier.

Local Approaches

Similar to many other local agencies, MnDOT and the Council have adopted similar tools and methodologies to estimate project and program-level emissions. The Council utilizes various methodologies for emission estimation in its RS project applications, including calculations and spreadsheet analyses based on VMT from traffic modeling analysis, as well as direct emission outputs from tools such as VISSIM, Synchro/SimTraffic, or HCM. Additionally, they use the regional ABM in conjunction with MOVES to estimate emissions for their TPP. However, it is worth noting that quantitative emissions analysis is not currently conducted for projects in TIP.

Recommend Methodologies for Estimating GHG Impacts

Building upon the review of the current Council GHG quantification practices and the review of other best practices, the consultant team has developed a set of recommended methodologies for estimating GHG emissions impacts. This section details research findings in response to specific questions about how to quantify the GHG impact of projects included in the RS and the TPP. **For TIP, it has been determined that emissions cannot be evaluated during the TIP phase due to lack of information reported. Project sponsors can either submit GHG estimates in line with established guidelines or propose their own methods.** TIP project construction emissions can be quantified using the MICE tool. Several guiding principles and key considerations have been established for the team to provide these recommendations:

- Balance effort against precisions and completeness of GHG impact estimates;
- Account for vehicle WTW emissions;
- Capture emission benefits throughout the entire project life span;
- Apply national best practices with regional-specific input parameters; and
- Provide recommendations to collect project-related information for future methodology refinement.

Transportation Policy Plan Recommendations

The section summarizes major outcomes from three TPP working group (WG) meetings between Council staff, MnDOT, WG members, and the consultant team. The primary goal of these meetings was to refine the methodologies for reporting GHG emissions in the TPP. This TPP section covers key points discussed in three main areas: induced demand, construction emissions, and electrification.

Induced Demand

Induced demand in transportation refers to the phenomenon where the expansion or improvement of transportation infrastructure leads to an increase in travel demand beyond what would have occurred without the intervention. Forecasting induced demand resulting from transportation projects is inherently challenging due to uncertainties surrounding future behaviors, land use changes, and other external factors.

TDMs are widely used in transportation planning to forecast future travel demand and assess the impact of various transportation and policy scenarios. In the Twin Cities, the Council has been developing and updating its TDMs, using them in TPP process for many years. The model has evolved from its traditional four-step model to a more advanced Activity-Based-Model (ABM). Traffic volumes and VMT outputs from the TDMs are available for different vehicle types at individual roadway link levels within the model limits. These outputs can be aggregated into corridors, subregions, and entire metropolitan areas. Aggregated VMTs from different scenarios are usually used for benefit-cost analysis in transportation improvement projects.

While TDMs can incorporate behavioral components that simulate how individual travelers respond to changes in transportation infrastructure, they often require lengthy computation runtime and can pose significant complexity in calibration and validation, as well as limited abilities in predicting future behavior and land use changes. Alternatively, the State Highway Induced Frequency of Travel (SHIFT) tool, developed by the Rocky Mountain Institute (RMI), is currently widely known for estimating induced demand in terms of VMT and emissions in transportation planning due to its ease of use [16].

The consultant team has conducted a sensitivity test to explore how induced VMTs would respond when using the ABM for freeway expansion projects in the Twin Cities metropolitan area. This analysis also aimed to assess the differences between results obtained from ABM and those from the RMI's tool – a much simpler estimation method. It is worth noting that the RMI's tool results are for freeways only, while the results from

the ABM are for all roadways in the model (functional class collector roadways and above). Nevertheless, below are the key findings regarding the overall induced VMTs from the ABM and the freeway results from the RMI tool:

- The metro wide yearly VMTs on freeways in the 2018 ABM, particularly around 5.6~5.7 million VMTs per freeway lane-mile, generally align with the results from the RMI tool (5.5 million VMTs per freeway lane-mile). However, the ABM shows varying elasticity factors for induced VMTs in response to freeway improvement projects. These elasticity factors are influenced by different corridors, land uses, transportation environments, and congestion levels over different years. Unlike the fixed factors of 1.0 for interstate and 0.75 for non-interstate highways in the RMI tool, they are dynamic and responsive to different corridor conditions. **Lower elasticity factors are recommended when using the RMI tool for induced VMT estimation.**
- While the current focus of induced VMT estimation for freeway improvement projects is on the freeways, the ABM modeling analysis indicates that **diverted VMTs from non-freeways are noticeable and should be considered depending on study purposes.** The ABM serves as a valuable tool and is recommended for evaluating comprehensive impacts of transportation improvement plans on both freeways and non-freeways.
- Induced VMTs display a non-linear relationship with the addition of lane-miles. Factors such as network congestion levels and project locations can influence induced demands.
- Project influence areas may vary over time due to network congestion changes. Researchers or engineers should exercise caution when defining the project influence area for induced demand analysis.
- Metro-wide induced VMTs in 2040, with higher congestion levels, are generally lower than those in 2018. The ABM modeling analysis shows that transportation improvements in more congested networks are anticipated to yield fewer induced VMTs. This is because the added capacity is initially absorbed by traffic in congested networks before addressing latent induced demand. Capacity improvement projects in free-flow corridors typically do not introduce induced demand if the land uses are not influenced by these improvements. It is worth noting that the current ABM modeling process used for this analysis doesn't include automatic feedback to land use plans. Therefore, there may exist a critical congestion threshold where project capacity improvements lead to the highest induced demand. As the corridor's congestion level surpasses this critical point, induced demand is expected to decrease in response to capacity improvement projects, indicating greater efficiency from the perspective of induced VMTs.

Construction Emissions

A significant point of the TPP discussion was the treatment of construction emissions. The TPP currently evaluates on-road emissions, while it is understood that construction emissions can reflect a high proportion of a project's emissions across the entire lifetime. Notably, construction emissions are often associated with non-road vehicles, equipment, and materials, making them distinct from the on-road sector and evaluation methodologies used to determine emissions.

The Minnesota Infrastructure Carbon Estimator (MICE) Tool

The current methodologies used to inform the Council TPP cannot produce estimates of the component parts of GHGs, especially as it pertains to construction, manufacturing, and disposal. However, construction emissions can still be quantified offline using the MICE tool, which is a Minnesota-specific version of the FHWA Infrastructure Carbon Estimator tool, incorporating local vehicle emissions rates, the capability for build/no-build analysis, and other enhancements [10]. The tool, maintained by MnDOT, aims to provide lifecycle estimates of energy and GHG emissions based on national emission and energy use factors for materials and construction activities. Projects of which constructions emissions can be quantified use MICE include bridges

and overpasses (new construction or reconstruction), bus rapid transit, culverts and storm sewers, light and heavy rail, lighting, parking, pathways (trails and sidewalks), roadways (including roadways resurfacing and reconstruction), and signage. Starting on January 7th, 2020, MnDOT committed to analyze GHG emissions from transportation projects as part of the environmental review process, which requires all eligible state and federally funded projects on MnDOT facilities to evaluate construction and maintenance emissions using MICE. Projects exempted from this requirement include those that have an Early Notification Summary Memo for the environmental document, those with a total construction cost of less than \$1,000,000, and those that cannot be quantified using the MICE tool.

FTA Evaluation of Construction Emissions for Transit Projects

In addition to initiative taken by MnDOT, the Federal Transit Administration's (FTA) has also studied the importance of considering construction emissions for project evaluation and GHG mitigation. The FTA published its 2024 Programmatic Assessment to quantify five such projects: light rail, street car, bus rapid transit, commuter rail, and heavy rail [17]. The results are based off 68 transit project scenarios modeled using the FTA GHG Emission Estimator tool. This study assessed both downstream and upstream emissions over the construction, maintenance, vehicle operative and emission displacement phases of these transit projects, inclusive of the properties as described in Table 8.

Table 8. FTA Estimator GHG Emissions Sources Included in Programmatic Assessment (2024)

Phase	GHG Emissions Sources Included
Construction	Transitway track construction Paving of separated rights-of-way Station construction Parking construction Catenary system (construction/copper)
Maintenance	Routine maintenance of transitway Routine maintenance of pavement Routine maintenance of vehicles
Vehicle Operations	Transit vehicle miles traveled (VMT)
Displaced Emissions	Automobile VMT Transit VMT

The FTA Estimator tool's primary data source for construction related emission factors and maintenance emission factors are from the FHWA's ICE tool. For vehicle operations, the Estimator uses Argonne National Laboratory's GREET model for available vehicle and fuel type combinations, including electric variants. The FTA Estimator also assumes a certain level of "ridership affect", which affects the estimated net GHG emissions associated with transit modes. Some of the assessment's findings include:

- Light rail projects, regardless of length, alignment, and number of stations, generally result in net reductions in annual GHG emissions over the minimum useful light rail lifespan of 50 years. Annual displaced GHG emissions due to light rail's "ridership effect" are greater than the GHG emissions from construction, maintenance, and vehicle operation phases.
- Bus rapid transit projects generally result in relatively low but net increases in annual GHG emissions over the minimum useful bus rapid transit lifespan of 40 years. The annual GHG emissions from construction, maintenance, and vehicle operation phases are greater than the GHG emissions displaced as a result of ridership effect.

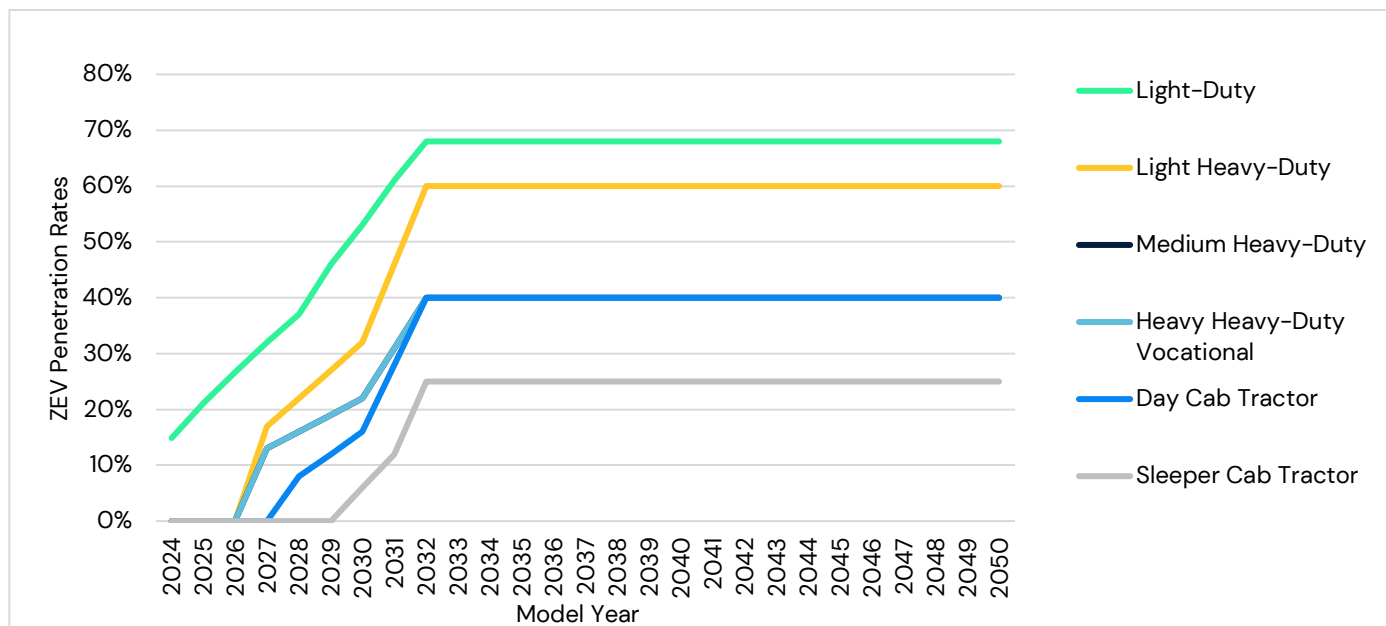
- The commuter rail projects result in annual net increases in GHG emissions over the minimum useful lifespan of 50 years. The annual GHG emissions from construction, maintenance, and vehicle operation phases are greater than the GHG emissions displaced as a result of the ridership effect.

A key takeaway from this discussion is the necessity for accounting for infrastructure-related construction emissions. **Given that the MICE tool is easily accessible for the Council, the project team recommended the Council staff start using MICE tool for project construction emissions quantifications (with same exemption terms applied).** While the fraction of construction emissions as compared to vehicle tailpipe or project lifecycle emissions may vary, it is critical to consider consistency in how construction emissions are treated across different project types. Regardless of the relative emissions between roadway and bicycle or pedestrian projects, if construction emissions are included in one type of project, they should also be included in the other to ensure consistent emissions estimates.

Electrification

To date, there is no methodology to account for vehicle electrification and zero-emission technology penetration in the TPP. EV penetration could be modeled using projections from different sources, such as the MOVES, which includes the new Multi-Pollutant and Greenhouse Gas emissions standards for light-, medium-, and heavy-duty (LD, MD, HD) on-road vehicles and engines for MY 2027 to 2032 [18, 19]. Note that the penetration rates do not specify or separate different ZEV technologies and battery-electric, plug-in hybrid electric, and fuel-cell electric technologies are all combined. While the regulations remain technology-neutral and performance-based, allowing each manufacturer to choose what set of emissions control technologies is best suited for them and the needs of their customers, the regulations did provide compliance scenarios that heavily rely on zero-emission technologies, as shown in Figure 2. Given the rules have been officially adopted by the EPA, the project team recommends the Council to consider including the new emission standards as a scenario for further emissions and compliance studies.²

Figure 2. ZEV penetration rates proposed under the EPA Multi-Pollutant and Greenhouse Gas Emissions Standards.



² EPA has recently announced reconsideration of the light-duty, medium-duty, and heavy-duty vehicle regulations that provided the foundation for the Biden-Harris electric vehicle mandate, introducing uncertainties to the EV market. More can be found at: <https://www.epa.gov/newsreleases/epa-launches-biggest-deregulatory-action-us-history>

Alternatively, the Council may also consider leveraging the U.S. EPA MOVES4 projections as a baseline to be modulated by Statewide Multimodal Transportation Plan (SMTP) [20]. The SMTP contains regional electrification targets which are based off the Next Generation Energy Act and the Paris Agreement. According to MnDOT, there were approximately 24,000 EVs registered in MN as of December 2021. More than 80% of EVs are registered within the major seven-county metro area, however, this reflects less than 1 percent of all vehicles registered in the state. There is a large discrepancy between the current number of EVs registered and the staggered targets for electrification, the first of which is 5% EV penetration by 2025. Reportedly, having 5-year targets out of the total number of all vehicles sold for 2030, 2035 and 2040 provides indicators to the shift in overall transportation sector GHG emission reduction. These goals can be applied against U.S. EPA projections or other local targets to produce wider ranges of change to the state's vehicle population.

For transit electrification, Metro Transit has made significant strides in electrification, currently operating eight battery electric buses. In 2023, Metro Transit received a Low or No Emission grant to procure 12 new battery electric buses and supporting chargers. In addition, the Council created their Zero-Emission Bus Transition plan, laying out their plans and goals to decarbonize their transit fleet [21]. Metro Transit has set a target for at least 20% of 40-foot bus replacements purchased between 2022 and 2027 to be electric. This approach is further evidenced by the fact that approximately 33.3% of all transit rides in 2023 were on electrically powered transit vehicles, including the Green Line and Blue Line light rail lines. Given the adaptability of MOVES, transit electrification targets specific to the region shall also be considered and quantified.

Regional Solicitation Recommendations

The RS projects are divided into five categories: electrification, transit, bicycle and pedestrian (bike/ped) facilities, transportation demand management, and roadways, as shown in Table 9. In addition to quantification methodologies, the memo also addresses several research priorities identified by the project team, along with WG members and Council staff. A total of five WG meetings, one for each category, were hosted by the project team during October – November 2023 to determine the primary focus for GHG quantifications and project types. Table 10 provides the suggested project life span for each RS project type.

Table 9. RS project types considered for GHG quantification methodologies.³

Category	Project Type
Electrification	EV Education and Outreach
	Public Charging Infrastructure
Transit	New or Expanded Transit Service
	Mobility Hubs
Travel Demand Management	Employee Commute VMT Reduction
	Shared Mobility Programs
Bicycle and Pedestrian Facilities	Pedestrian Facilities
	Multi-Use Trails and Bicycle Facilities
Roadways	Intersection Delay Reductions
	Corridor Speed Increases

³ For project types that may include a variety of applications and project components, such as Safe Routes to School Infrastructure, Traffic Management Technologies, and Spot Mobility and Safety, it is rather difficult to use a single methodology for quantification. Therefore, they are not included in this document.

Table 10. Default lifetime for the RS projects.

Regional Solicitation Project Type	Default Project Lifetime (Years)	Source
Transportation Demand Management – Employee Commute VMT Reduction	4	U.S. Bureau of Labor Statistics (BLS) [22]
Transportation Demand Management – Micromobility	5	Assumptions used by the MnDOT Carbon Reduction Strategy [23]
Traffic Management Technologies	7	U.S. Department of Transportation (USDOT) Benefit–Cost Analysis Guidance [24]
Electrification Education and Outreach –Light Duty Personal, Transportation Demand Management – Carshare, Transportation Demand Management – Carpooling/Vanpooling	8	Federal Transit Administration (FTA) Useful Life Benchmarks [25]
Public Infrastructure Deployment	10	Assumptions used by the MnDOT for the National Electric Vehicle Infrastructure (NEVI) Formula Program ⁴
Electrification Education and Outreach –Medium and Heavy–Duty Fleets and Commercial, Transit Expansion – New or Expanded Transit Service	14	FTA Useful Life Benchmarks
Spot Mobility and Safety, Strategic Capacity, Transit Hubs, Transit Expansion – Bus Rapid Transit Conversion, Transit Modernization, Bike & Ped Facilities	20	USDOT Benefit–Cost Analysis Guidance

Transit

Projects that add or expand bus service in the metropolitan area can improve the transit coverage area, provide transit service to more residents, and reduce trips and emissions from single-occupancy vehicles (SOVs). The proposed methodology would require ridership forecasts from project applicants and consider the percentage of new transit users who would have driven if transit options were unavailable (referred to as adjustment factor). Since new or transit services often involve added travel from transit buses, new emissions from added transit are also reflected. Rather than relying on terminal-to-terminal distances, it utilizes the average distance of transit trips to calculate the reduction in VMT for each trip, with data provided by Metro Transit as summarized below.

Table 11. Default trip length and adjustment factor by service type, data provided by Metro Transit.

Route Type	Average Trip Length (mi/trip)	Adjustment Factor
Bus Rapid Transit	13.2	0.62
Commuter Express	16.3	0.705
Commuter Rail	38.8	0.867
Core Local	12.5	0.62
Light Rail Transit	11.4	0.685
Suburban Local	9.1	0.62
Support	9.2	0.62

⁴ Via communication with MnDOT staff.

Based on the WG discussion, project applications to develop multi-modal mobility hubs around transit stations with pedestrian/bicycle facilities and car-sharing services are likely to increase in the upcoming years. A mobility hub is a carefully designed space that integrates various forms of transportation and related services to facilitate seamless and efficient movement of people. Key elements typically found in a mobility hub include:

- **Public Transit Access:** This is a core feature of a mobility hub, providing easy access to multiple modes of public transportation like buses, trains, trams, or subways.
- **Bike-Sharing Stations:** Facilities for renting bikes, often including electric bikes, which provide an eco-friendly and healthy transportation option.
- **Car-Sharing Services:** Availability of shared cars that can be rented for short periods, reducing the need for personal vehicle ownership.
- **EV Charging Stations:** Infrastructure to support the charging of electric vehicles, promoting the use of environmentally friendly transportation.
- **Pedestrian-Friendly Design:** Emphasis on safe and accessible walkways, promoting walking as a viable mode of transportation.

Ideally, the best way to assess the VMT and emissions impact of mobility hubs is a two-step approach: first, measure the community's VMT reduction from the mobility elements derived from the Council's Mobility Hubs study [26]; then calculate the VMT displacement from increased transit ridership due to better transit access provided through these elements. However, the estimation of transit ridership involves many variables and often requires a large amount of data and extensive regional travel demand modeling before any conclusion can be drawn. As an emerging project type, there is little information on how mobility hubs can affect or improve transit ridership. One recent study suggests that a 10% growth in population covered by transit using first & last mile connection can increase transit usage by 5.9% [27]. However, the impact of ridership can change drastically depending on population density and service types. Therefore, the project team proposes this simplified alternative to the methodology the team has also provided an alternative to estimate community VMT reduction from mobility and mobility-to-transit combined, avoiding the need to assess ridership impact. The VMT reduction factors are adjusted from a study conducted by the Orange County Transportation Authority [28], and community VMT is calculated using the buffer distance of 1 mile for the population within the service area. **Further discussion is required to refine the methodology for estimating community VMT and assessing VMT reduction potential of various mobility elements.**

Travel Demand Management

The methodologies for transportation demand management projects are classified into two categories based on the main audience and affected parties: commuter trip reduction and shared mobility initiatives.

Employers or companies typically sponsor commuter trip reduction projects to decrease their employees' commute trips using various modes. For this type of project, it depends on project sponsors to assess the reduction in the number of commute trips and the recommended methodology considers the differences in average commute distances throughout the region, based on the Council's latest TBI data.

The concept of shared mobility applies to any project that encourages the shared use of a vehicle, bicycle, or other mode, including carshare, bike/electric bike share, scooter/electric scooter share, carpooling, vanpooling, on-demand ride, ride-hailing, etc. Note that transit expansion projects are considered under the transit category, and thus not included in this memo. The recommended methodology for shared mobility is adapted from CARB to assess shared mobility emission reduction potential from an estimate of displaced auto VMT by mobility service type. This methodology also accounts for any new VMT, and WTW GHG emissions associated

with the shared mobility service. Because high variances in trip distance from different mobility services have been observed and shared mobility programs are not solely targeted at commute trip reduction, the project team recommends applying trip distance by project type instead of using the same commute trip distance as in the commuter trip reduction methodology. In addition, the methodology also accounts for deadhead miles for ridesharing or carsharing services.

Bicycle and Pedestrian

The recommended methods for bike/ped projects mainly use elasticity models to estimate the percentage of current vehicle trips that can be converted to bike/ped trips. The approach requires an annual average daily traffic (AADT) estimate on the road next to or parallel to the new facilities. Facilities built close to key destinations⁵ can earn additional credits. The Council plans to create map layers to inventory key destinations in order to standardize the credits given to new facilities. Additionally, the methodology provides higher credits for protected bike lanes to address the safety concerns of cyclists who are "Interested but Concerned".

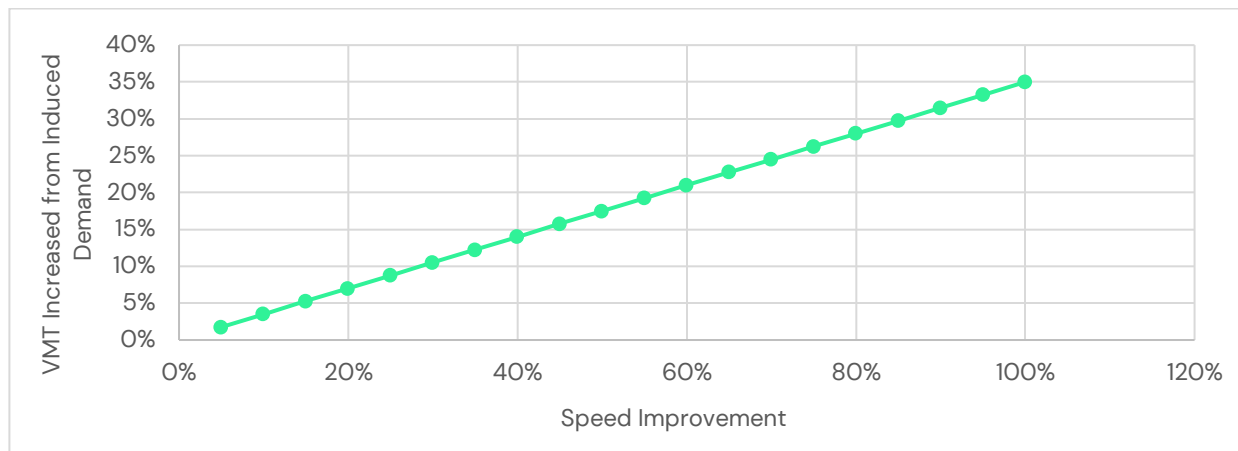
Roadways

The GHG quantification of roadway projects primarily follows the existing methodology using Synchro and SimTraffic. The emission factors related to idling and speed-dependent running exhaust have been revised to reflect the latest emission standards in the new methodology.

A new component has also been added to evaluate the induced demand associated with improved corridor speeds. Many RS Roadways projects focus on operational improvements that increase effective capacity, without physical lane additions. These projects can be expected to cause induced vehicle travel to the extent that they improve travel speeds in a corridor. A study from the 2000s investigated the change in travel miles as a result of reduced travel time, which is correlated to speed increases [29]. The study collected data from over 27,000 households, and the findings indicate that travel-time elasticities range from -0.350 to -0.582.

This implies that a 10% reduction in average household travel time will result in at least 3.5% increase in annual household VMT. The following assumptions, as shown in Figure 3, can thus be made for estimating induced vehicle travel from speed increases.

Figure 3. Induced VMT as a function of speed increases.



⁵ Eligible key destinations encompass banks, post offices, childcare facilities, grocery stores, medical centers, office parks, pharmacies, places of worship, public libraries, public parks, schools, universities, and colleges.

Electrification

This new project category includes two project types: public charging infrastructure deployment, and EV education and outreach. The methodology compares GHG emissions of internal combustion engine vehicles to those from EVs, without considering any VMT shifts from EV adoption. The approach considers emissions from the upstream generation of electricity, recognizing that EVs are not entirely "zero-emission". The existing approach depends on regional average, calculated from limited data points. The consultant team suggests that the Council should continue collecting data to refine these default values, particularly focusing on:

- Current and upcoming charging infrastructure projects to understand regional charger usage, with a particular emphasis on the active in-use hours for different types of charging stations (Level 2 vs. Direct Current Fast Charging).
- Outcomes from outreach events to understand how effectively these events encourage drivers to transition to EVs.
- Future methodological and data enhancements could directly address medium- to heavy-duty EVs.

Review and Update GHG Inventory

A critical aspect of estimating the impact of GHG emissions for future projects is to establish a robust understanding of the GHG baseline and develop a business-as-usual (BAU) forecast. As mentioned earlier, the SP tool has already been used by the Council to estimate current and future regional GHG emissions from on-road transportation. To gain a comprehensive understanding of the SP tool and BAU forecasts, the project team examined its code repository, the Council's Priority Climate Action Plan (PCAP) and Climate Pollution Reduction Grants (CPRG) inventory [30], and various other pertinent technical documents.

Strategy Planning Tool GHG BAU Forecast

The SP tool has been used to estimate transportation sector emissions for the Twin Cities metropolitan area, reflecting counties found in the Minneapolis – Saint Paul, Minnesota urbanized area. Transportation sector emissions considered include both direct and embodied (indirect) emissions, such as manufacturing, maintaining vehicles, and infrastructure. Applicable emission reduction strategies include land use, vehicle electrification, mode shift (use of public transit), road pricing, and parking policy. Previously, a BAU forecast was developed based on a set of standard forecasts for inputs such as passenger miles traveled (PMT), VMT, existing vehicle stock, new vehicle sales, average vehicle occupancy, fuel efficiency, and population. The impact of different emission reduction strategies was estimated using changes in combustion PMT as a proxy. While the impact of transit usage and EV adoption on passenger miles are relatively straightforward to estimate, multiple elasticities models were established between different intervention adjustments, as shown in an earlier example for the City of Minneapolis (Table 7).

The SP tool is written in the R programming language and has cloned the SP tool repository to extend and refine the existing GHG estimation methodology. The SP tool's R directory organizes building, land use, transportation, and walk/bike modes into distinct modules. The team began by analyzing the transportation modules required inputs (listed below) which are used to calculate VMT, indirect and direct GHG emissions, and cost. Summarized below are the required inputs needed to run the transportation module.

- **Passenger Modes Data (5-year increments 2015–2050):** The passenger modes dataset is the baseline for vehicle miles travelled. The powertrain types included in the dataset are gasoline, diesel, hybrid electric–gasoline, plug-in hybrid electric gasoline, and battery electric. To attain the vehicle stock captured in this dataset, Sustainable Healthy Cities Network (SHCN) used the Market Acceptance of Advanced Automotive Technologies (MA3T) along with outputs from the regional TDM. State level vehicle stock estimates were then downscaled to cities, townships, and unorganized territories (CTU) by population.
- **Freight Modes Data (5-year increments 2015–2050):** The freight modes dataset supplies the same data as the passenger modes data and is similarly aggregated to CTU in 5-year time stamps. To estimate vehicle stock distribution, MA3T was used with a factor of 2/3 applied to the BEV stock proportion to account for lower market penetration in long-haul freight.
- **Electric Scenario (default 'ER'):** The standard electricity scenario provided is labeled ER in the module and reflects the Xcel energy reference scenario electricity generation mix.
- **AEO Scenario (default 'REF'):** The AEO scenario captures various global uncertainties in key energy markets and are incorporated into the tool as variations in fuel economy and VMT by mode.
- **CTU and Transit Market Areas (TMA):** The tool breaks up vehicle miles travelled and stock by CTU, which identifies all cities, townships, and Census Bureau-defined unorganized territories in Minnesota. VMT and Vehicle stock are downscaled to CTU by population. CTUs are identified by the highest TMAs.

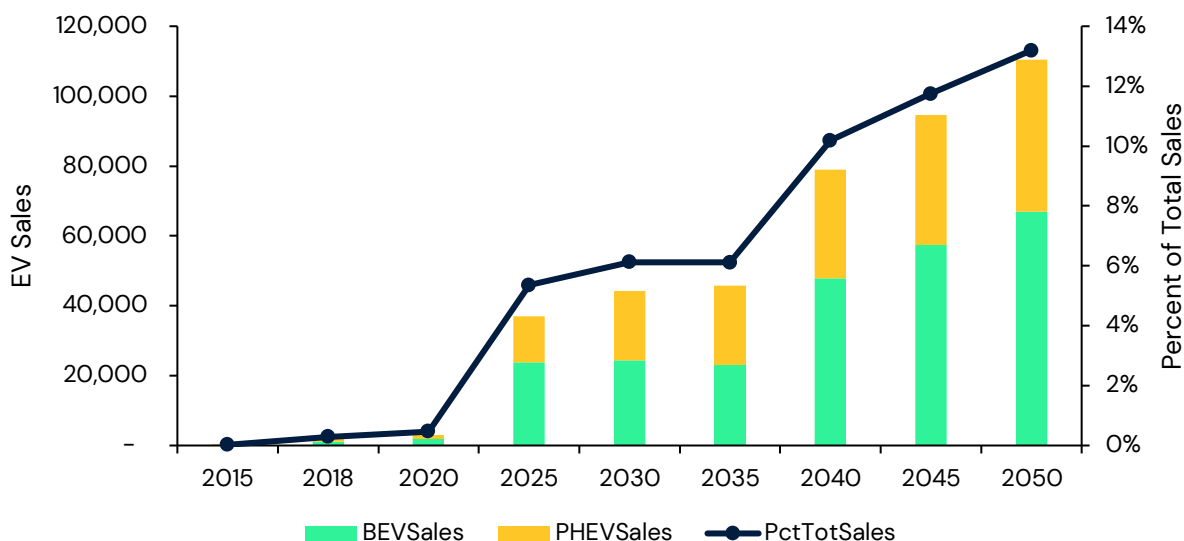
The SP tool breaks up vehicle stock into two modes: Passenger and Freight. The fleet data utilizes VMT information from the regional TDM, which is subsequently downscaled to individual CTUs. This VMT data is utilized to compute direct emissions based on the specific mode and fuel type. For non-electric vehicles, the resulting value represents tailpipe emissions. However, for electric fuel types, the value represents the equivalent emissions per kilowatt hour (kWh), adjusted by the percentage of the grid that has been decarbonized. If the entire grid is decarbonized, there are no emissions associated with EVs. Considering the discrepancies in current GHG scope between EVs and non-EVs, the consultant team suggests the Council **account for well-to-pump emissions (e.g., fuel production, distribution, and delivery) of non-EV fuels along with tailpipe emissions for an accurate and fair comparison to EVs.**

The SP tool's transportation module documentation describes, at a high level, how vehicle electrification is considered when running a defined transportation scenario (Table 12). To illustrate, Figure 4 shows the passenger vehicle battery-electric (BEVs) and plug-in hybrid electric (PHEVs) sales projections, respectively. **One consideration to improve the current EV sales projections are to use those consistent with the U.S. EPA's EV sales curves or MOVES5 forecasts, which provide different projections for light-, medium- and heavy-duty vehicles.** The U.S. EPA EV sales curves escalate EV sales projections between 2027 through 2032, which are assumed constant or can be adjusted based on state goals.

Table 12. Transportation module electrification intervention workflow.

Intervention	Two-line description in simple language	Calculation Method
B1. Increase private EV sales to 20% HEV, 20% PHEV, and 60% BEV of all vehicle sales by 2040.	EVs are more energy efficient than internal combustion engine (ICE) vehicles and will further reduce emissions as the grid decarbonizes. The model assumes 6% of vehicles are hybrid electric (HEVs) with fuel economy of 65.6 mi/gal, 4% of vehicles are PHEVs with fuel economy of 57.1 mi/gal in gasoline-mode and 3.1 miles/kWh in electric-mode, and 6% of vehicles are BEVs with fuel economy of 3.6 mi/kWh in baseline projection by 2040. The grid emissions factor in 2040 is 0.44 kg CO ₂ eq/kWh based on the Xcel Energy reference scenario.	A baseline forecast is based on MA3T data. The Council region is disaggregated into a set of smaller geographic regions. Population, vehicle ownership, and other input data are taken from census and travel survey sources. These forecasts use 2015 as a base year for vehicle stocks and sales and compared these results against control totals for 2020. Vehicle electrification changes the vehicle sales and stocks. These changes are then translated into changes in VMT and GHG emissions (direct and indirect) by powertrain type through the VMT equation.
B2. Phase out biodiesel buses from fleet by 2035 and HEV bus sales by 2050.	Battery-electric buses (BEBs) are more energy efficient than biodiesel buses and will further reduce emissions as the grid decarbonizes. The model assumes that all biodiesel buses are phased out by 2035. The bus fleet is 31% hybrid-electric and 69% battery-electric by 2040.	Changes to the bus fleet produce changes in the powertrain mix of the bus stock. These changes then translate into changes in VMT and GHG emissions (direct and indirect) by powertrain type through the basic VMT equation given above.

Figure 4. Passenger EV sales projections



The SP tool's R directory uses total vehicle sales and GHG emission factor data to calculate transportation embodied emissions. This data is segmented by model year and fuel type, such that the output returns the embodied emissions results for ICEVs, HEVs, BEVs, and PHEVs. The embodied emission factors are obtained from the EPA's 2014 Emission Factors for Greenhouse Gas Inventories, which provide data on embodied emission factors categorized by vehicle fuel type [31]. The EPA has recently updated its GHG Emission Factors Hub, with the most recent update in January 2025 [32]. The team recommend that the Council should align its underlying assumptions with the latest EPA update. It is also possible to calculate fleet embodied emissions using emission data from the Greenhouse gases, Regulated Emissions, and Energy use in Technologies (GREET) model [33], which has a more comprehensive methodology and workflow for calculating emissions from vehicle manufacturing. GHG emissions from EV manufacturing are higher than those from ICEV, but the disparity is projected to decrease in the future. Thus, it is more effective to base embodied emissions on upstream fuel pathways, as calculated in GREET, to reflect future trends.

Comparison with the CPRG GHG BAU Forecast

The data used to compile the CPRG inventory transportation sector comes from [StreetLight](#) Volume Data. For passenger vehicles, StreetLight Volume estimates of volume and trip length were used to create the vehicle miles traveled baseline. For commercial vehicles, the StreetLight Index was used to measure traffic volume, along with an AADT calibration value. Data from MnDOT and the Wisconsin Department of Transportation (WisDOT) were also used for calibration and validation. All of the data sources used to estimate emissions for the transportation sector are in Table 13.

Table 13. CPRG transportation data sources for emissions calculations

Data Source	Use
StreetLight Data	Data on travel patterns, traffic volume (number of vehicles), and average trip length for passenger and commercial vehicles. Used for origin-destination analysis and estimating VMT.
MnDOT	Data on county-level VMT, AADT, and vehicle distribution by weight. Used for calibration and validation of StreetLight Data.

WisDOT	Data on county-level VMT, AADT, and vehicle classification data. Used for calibration and validation of StreetLight Data.
EPA MOVES	Emissions rates based on regional vehicle characteristics. Used for calculating vehicle emissions.
TBI	Household survey data on travel behavior. Used to validate average trip lengths and regional fleet characteristics.

To estimate VMT, the Council calculated the number of vehicles and average trip length in miles for all origin-destination pairs and vehicle types. Unlike SP's baseline fleet stock, the CPRG inventory does not account for public transportation or mass transit. The CPRG aggregates annual emissions by county for the Twin Cities metropolitan area.

To compare the baseline VMT and emissions in the SP tool to the CPRG inventory, the consultant team used the tool's BAU data for 2020. We aggregated individual CTU projections to the county level, then filtered the data annually and included only passenger emissions, as shown below.

Table 14: Comparison of VMT for the base year in the SP tool and CPRG GHG inventory

County	SP VMT (Mile)	CPRG VMT (Mile)	Difference (Mile)	Percent Difference
Anoka	3,024,270,332	2,832,786,225	-191,484,107	6.5%
Carver	1,294,326,030	813,325,102	-481,000,928	45.6%
Dakota	3,984,504,076	3,303,557,677	-680,946,400	18.7%
Hennepin	13,278,836,575	7,675,961,096	-5,602,875,479	53.5%
Ramsey	5,114,286,225	3,525,944,877	-1,588,341,348	36.8%
Scott	1,822,711,067	1,313,168,658	-509,542,410	32.5%
Washington	3,136,327,575	2,422,839,340	-713,488,235	25.7%

Table 15: Comparison of GHG emissions for the base year in the SP tool and the CPRG GHG inventory

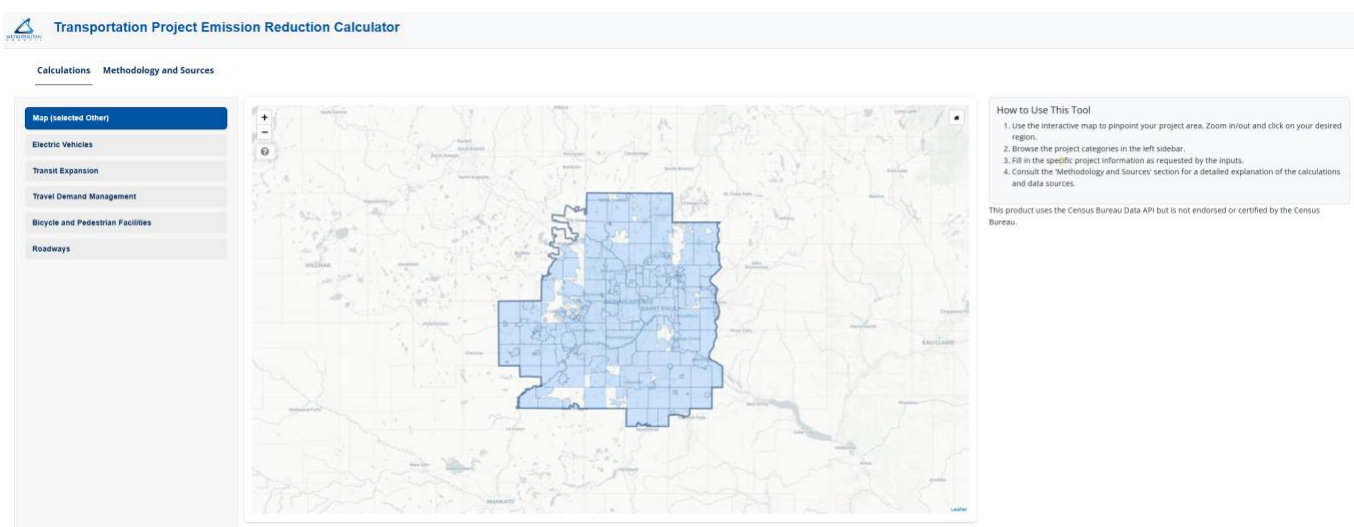
County	SP Emissions (MT CO ₂ e)	CPRG Emissions (MT CO ₂ e)	Difference (MT CO ₂ e)	Percent Difference
Anoka	978,175	955,750	22,425	2.3%
Carver	346,273	277,586	68,687	22.0%
Dakota	1,176,491	1,077,718	98,773	8.8%
Hennepin	4,256,910	2,578,205	1,678,704	49.1%
Ramsey	1,701,741	1,186,822	514,919	35.7%
Scott	560,780	439,319	121,461	24.3%
Washington	982,189	823,507	158,682	17.6%

Considering the distinct data sources for each model, it's not surprising that the two inventories display different results. The project team recognizes that the SP tool is meant for long-term planning, whereas the CPRG inventory is a single-use calculation. **However, the comparison highlights the importance of regularly consulting new sources and updating the inventory as vehicle activity, emission standards, and EV adoption evolve, while also cross-referencing default vehicle activities and trip characteristics with transportation big data analytics.**

Implement Recommended Methodologies for Example Projects and Programs

To support the implementation of the Council's GHG quantification process, the consultant team has developed an R-based Transportation Project Emission Reduction Calculator, illustrated in Figure 5. This calculator is intended for use in future RS processes to calculate GHG emissions impact according to the recommended methodologies provided by the team. The team has tested the proposed GHG quantification methodologies for a sample of projects and scenarios. For Roadways, the team has also compared the Synchro report output to that of the proposed methodologies and the Transportation Project Emission Reduction Calculator. The project team also assessed the availability of data from completed applications that could be used with the FHWA CMAQ Calculator Toolkit.

Figure 5. The Council's Transportation Project Emission Reduction Calculator⁶



Transportation Project Emission Reduction Calculator

The Transportation Project Emission Reduction Calculator is an easy map-based tool that can be used to quantify RS project GHG emissions impact, the tool uses the Census Bureau Data API to extract necessary information for project quantifications. To start quantifying GHG emissions, users need to first use the interactive map on the front page to identify the project area. This is critical for estimating the residential population for Mobility Hubs and aligning the CTU with designated land use categories that may have varying fleet mixes and community VMT assumptions. Once the location has been identified, users should proceed to navigate through the project categories located in the left sidebar to select the most suitable category for their projects. As shown in Figure 6, each project requires users to fill in the specific project information as requested on each page. Some input fields also have pre-populated default values provided by the consultant team based on existing data. However, it is recommended that users gather input suitable for their specific projects to obtain more accurate results. The tool also includes a 'Methodology and Sources' section for a detailed explanation of the calculations and data sources. The results are presented both annually and

⁶ Accessible via: <https://metcouncilcf.shinyapps.io/transport-emission-shiny-app/>

cumulatively, including impact on vehicle activities (e.g., VMT), GHG emissions, and social costs of carbon (SCC). These results can also be downloaded as a .CSV file directly using the integrated button function.

Figure 6. An example of project-level GHG calculation from the Transportation Project Emission Reduction Calculator

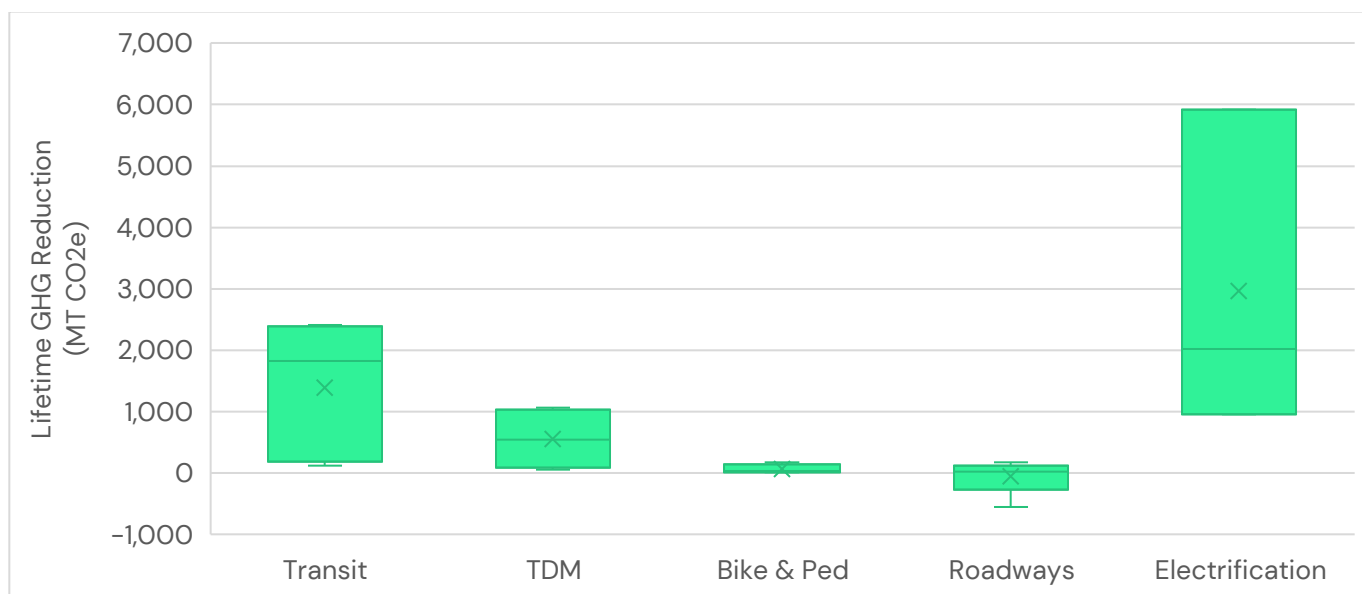
The screenshot displays the 'Transportation Project Emission Reduction Calculator' interface. On the left, a sidebar lists project categories: Map (selected Other), Electric Vehicles (highlighted), Transit Expansion, Travel Demand Management, Bicycle and Pedestrian Facilities, and Roadways. The main area is divided into two tabs: 'EV Education and Outreach' and 'Public Infrastructure Installation'. The 'Public Infrastructure Installation' tab is active, showing input fields for 'Type of Charger Installed' (DCFC selected), 'Location' (Brooklyn Park), 'Number of Chargers' (10), 'Annual Hours Available' (8,760), 'Project Lifetime (in years)' (10), 'Charger Power Level (kW)' (150), 'Year' (2024), 'Utilization Rate (%)' (5), and 'Percentage of EV in Regional Fleet (%)' (5). Below these fields is a 'Selected Community Type: Suburban' dropdown. At the bottom, a 'Public Infrastructure Results' table shows VMT Reduction (Miles), GHG Reduction (MT CO₂e), and Social Cost of Carbon Reduction (\$) from 2024 to 2033. A 'Download CSV' button is located to the right of the table.

Year	VMT Reduction (Miles)	GHG Reduction (MT CO ₂ e)	Social Cost of Carbon Reduction (\$)
2024	1,649,269	433	90,024
2025	1,649,269	448	94,982
2026	1,649,269	469	100,743
2027	1,649,269	491	107,423
2028	1,649,269	536	119,438
2029	1,649,269	554	125,232
2030	1,649,269	557	126,045
2031	1,649,269	550	126,722
2032	1,649,269	546	129,332
2033	1,649,269	537	129,424

The consultant team recommended to monetize GHG emissions benefits using the SCC, a measure that is often required by the U.S. EPA and other federal agencies to demonstrate climate impacts of their rulemaking efforts. SCC estimates the long-term damage from GHG emissions, including impacts on agricultural productivity, human health, property damages and changes in energy system costs. The consultant team used the latest SCC values from the EPA's Report, where SCC values are provided annually between 2024 to 2050 [34]. The EPA SCC provides three different discount rates using the Ramsey (1928) discounting formula, and the current tool uses a 2.0% discount rate. Multiplying the change in emissions for a future year (MT CO₂e/year) by the discounted SCC value (\$/MT CO₂e) for that year yields the monetized value of future emission changes from the perspective of that year. The total present value of benefits from an action or measure is thus the sum of the discounted, monetized values for each year the measure produces emission changes.

The team has selected a series of sample projects from the 2022/24 RS applications to use the Transportation Project Emission Reduction Calculator to evaluate project GHG impact. As shown in Figure 7, electrification projects generally lead to higher benefits compared to other categories, while bike/ped projects result in the lowest emissions benefits due to the shorter vehicle trips that can be displaced by these projects. Roadways projects are trickier, as some may lead to "negative" impact because induced vehicle travel. As previously mentioned, reduced travel time and increased corridor speed are likely to induce additional travel along the same corridor, thereby increasing vehicle activity within the project area and increasing GHG emissions (i.e., negative GHG benefits). Although standard transit expansion can achieve GHG reductions comparable to other RS project categories such as transportation demand management and electrification, large mobility hubs offer significantly greater potential for reducing GHG emissions than the others.

Figure 7. RS sample project GHG impact – cumulative impact (MT CO₂e)



Roadways Project Quantification Comparison

Quantifying roadways projects is the most challenging among all, due to the complex nature of traffic patterns and activities, along with associated secondary impacts such as induced demand. To evaluate the reliability of current quantification tools and methodologies, the team has selected recent RS samples and employed various tools and data sources to calculate GHG emissions for comparison purposes. The examples tested include (a) an applicant's proposed roundabout to replace an unsignalized T-intersection, (b) an applicant's proposed roadway four-lane expansion project for Highway 5, and (c) an applicant's roadway reconstruction project using a roundabout.

Overall, it is straightforward to use conditions reported by Synchro to input into the Transportation Project Emission Reduction Calculator to obtain emissions data for Intersection Delay Reductions projects. For Corridor Speed Increases projects, speed is not explicitly reported in the current applications, which makes it more challenging to estimate the impact of induced vehicle travel and emissions. The team has also noticed inconsistencies between Synchro results and Measure A: Congestion Reduction/Air Quality data in the current application form. Without details on the methodology or underlying assumptions, we cannot verify the accuracy of claimed benefits from Measure A: Congestion Reduction/Air Quality information. Additionally, some applications only report build conditions without providing baseline conditions, making it impossible to compare and calculate emission differences.

Applications that provide sufficient data through project descriptions and Synchro output were also input into the CMAQ tools to better understand GHG emissions. Given that CMAQ estimated its own traffic congestion and delay, a measured disparity between the Transportation Project Emission Reduction Calculator and estimations using Synchro report and CMAQ tool can be observed. The comparable magnitudes between the Transportation Project Emission Reduction Calculator and the CMAQ results provide a certain level of confidence in the recommended methodology.

For comparison purposes, the consultant team has also analyzed the emission estimates from Synchro against those from SimTraffic. The emission estimation in Synchro is based on established emission factor

models/formulas from HCM. It provides macroscopic fuel and emission estimates based on aggregated traffic flow data. The emissions include CO, NOx, and VOC and the emission factors can be adjusted. Emission estimates in SimTraffic are more detailed as it simulates the movements of individual vehicles based on car-following and lane change models. It captures variations in speed, acceleration, and idling times. These detailed vehicle movements are then used to calculate fuel and emissions, including CO, NOx, and VOC, based on factors for different vehicle fleets.

Synchro uses static emission factors that may not account for all real-world variations, while SimTraffic can incorporate dynamic emission factors that reflect changes in vehicle operations. Synchro model emission results are suitable for high-level planning and optimization of traffic signals and intersection performance analysis, while SimTraffic model results are deal for detailed operational analysis and scenarios where understanding the impact of vehicle dynamics is crucial. Significant discrepancies are often observed in the fuel usage and emission estimates between Synchro and SimTraffic modeling results.

Similar discrepancies are often observed in the measures of effectiveness (MOEs) in traffic operations analyses between the two models. Reporting MOE outputs for Synchro is relatively straightforward once it is set up. The SimTraffic modeling process starts with a Synchro model but usually requires additional calibration to accurately reflect driver behavior, vehicle types, and traffic conditions before reporting MOE outputs. The calibration can sometimes be simplified, or it can be a labor-intensive process, depending on project purposes. It may also take significantly more time to process SimTraffic outputs than Synchro outputs.

In summary, while Synchro provides a quicker and broader estimation of emissions and traffic operations based on average conditions, SimTraffic offers a more detailed and dynamic analysis by considering individual vehicle movements. **The choice between the two depends on the level of detail and effort required and the specific objectives of the traffic and emission analysis.** The team also strongly recommend projects sponsors document both build and no-build scenarios when reporting Synchro or SimTraffic results for comparison purposes.

Conclusion and Next Steps

This study provides a detailed overview of the current methodologies used by the Council to evaluate GHG emissions in the current TPP, TIP and RS planning processes. It also reviews the best practices for assessing GHG emissions from transportation projects, including walking, biking, transit, highways, transportation demand management, and electrification, across other government agencies. The study also provides recommendations for the Council to develop and implement reliable methods to assess the GHG impact of these projects. It also emphasizes the need for continuous improvement and refinement of GHG quantification methodologies to ensure accurate and reliable assessments of transportation projects' emissions impacts. By implementing these recommendations, the Council can better prioritize investments and make informed planning decisions that contribute to the region's sustainability goals.

Beginning in Spring 2025, a Regional Solicitation Special Issue Working Group for Climate will be established to further explore this topic area and integrate methods into project scoring. The R-based Transportation Project Emission Reduction Calculator tool should be employed and included in the ongoing application process to assess projects. In developing the tool, the team has noted that some projects lack region-specific data and may necessitate data reporting from funded initiatives to better understand how the current elasticity or methodology applies to local projects. These projects include mobility hubs, EV education and outreach, and the deployment of public EV charging infrastructure.

For roadway projects, there are notable discrepancies among Synchro, SimTraffic, and FHWA's CMAQ results. To further refine the methodologies and quantify Roadways project impact, the Council may consider using a lookup table by project type (e.g., interchange, lane expansion) and scaling parameters (e.g., AADT, level of congestion reduction) in future methodologies updates. In addition to the current suite of tools used for Roadway projects, evaluating tools like VISSIM or AIMSUN through pilot studies could be beneficial.

Adopting the EPA MOVES model as the standard for vehicle operational emissions calculations would be an important step forward, as it accurately reflects current and projected fleet technology. The Council may collaborate with MPCA to gather regional fleet data for more precise emissions modeling results from MOVES. In the short term, the Council could apply MOVES vehicle technology penetration rates to current software outputs during post-processing. In the longer term, exploring direct links to more advanced simulation tools could offer even greater precision. The emission rates from MOVES could be integrated into both TPP and SP tool development, as well as the emission factors used in the Transportation Project Emission Reduction Calculator.

The concept of induced demand should be further examined, not only in the context of highway capacity expansion as currently addressed in the ABM. Research has shown that the addition of lane miles to local arterials and reduced travel time can also impact vehicle activities. The currently proposed elasticity method may help in recognizing these impacts. In the long term, the Council needs to start collecting data from funded projects to track actual outcomes and changes in indicators such as AADT and further refine and the regional sensitivities within the elasticity model. Exploring deeper integration with the Council's regional TDM/ABM capabilities holds great potential for capturing long-term induced demand and land-use feedback, offering a more holistic view for major projects or corridor studies. For emissions impact quantifications, the Council must ensure that induced demand is accurately integrated and accounted for.

For TPP and TIP processes, which often involve large-scale roadway and facility infrastructure projects, construction emissions can be quantified using MICE tool. The Council may consider establishing a project

threshold similar to MnDOT to begin engaging people and training them to use the tool, with a long-term plan to integrate it into the prioritization and evaluation processes.

A tiered analysis framework could also serve as a practical approach to balance accuracy with the burden placed upon funding applicants, project sponsors, and Council staff. By categorizing projects based on their impact level or investment needs, the framework could optimize the use of available tools and methodologies. For lower-impact projects, classified as Tier 1, simpler tools such as the R-based Transportation Project Emission Reduction Calculator or standardized factors could be employed to streamline the analysis. This approach can ensure efficiency while maintaining an adequate level of estimates required for decision-making. On the other hand, more complex or high-impact projects, considered Tier 2, could include detailed analysis methodologies such as MOVES, advanced simulation techniques, and nuanced induced demand calculations. While these analyses may require a higher level of effort, they provide the necessary precision to inform decision-making. A tiered framework could significantly simplify the overall process, ensuring that the level of effort is proportional to the project's scale and impact.

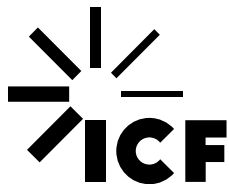
The continued partnership with MnDOT, MPCA, and local partners is crucial. Minnesota is one of the few front runner states who has established a regulatory framework and guidelines for quantifying the impact of project-level and planning-level GHG emissions and induced demand. Strengthening data-sharing agreements and collaboration platforms will support these analytical enhancements.

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